

As the time available for an energy measurement increases, the inherent uncertainty in the result decreases. In Section 9.5, we will apply this version of the uncertainty principle to the sharpness of spectral lines.

Example 5.4.2. Imagine an electron confined within a region of space the size of a hydrogen atom. We can estimate the minimum speed and kinetic energy of the electron using Heisenberg's uncertainty principle. Because we know only that the particle is within an atom-size region of space, we can take $\Delta x \approx a_0 = 5.29 \times 10^{-11}$ m. This implies that the uncertainty in the electron's momentum is roughly

$$\Delta p \approx \frac{\hbar}{\Delta x} = 1.98 \times 10^{-24} \text{ kg m s}^{-1}.$$

Thus, if the magnitude of the momentum of the electron were repeatedly measured, the resulting values would vary within a range $\pm \Delta p$ around some average (or *expected*) value. Since this expected value, as well as the individual measurements, must be ≥ 0 , the expected value must be at least as large as Δp . Thus we can equate the minimum expected value of the momentum with its uncertainty: $p_{\min} \approx \Delta p$. Using $p_{\min} = m_e v_{\min}$, the minimum speed of the electron is estimated to be

$$v_{\min} = \frac{p_{\min}}{m_e} \approx \frac{\Delta p}{m_e} \approx 2.18 \times 10^6 \text{ m s}^{-1}.$$

The minimum kinetic energy of the (nonrelativistic) electron is approximately

$$K_{\min} = \frac{1}{2} m_e v_{\min}^2 \approx 2.16 \times 10^{-18} \text{ J} = 13.5 \text{ eV}.$$

This is in good agreement with the kinetic energy of the electron in the ground state of the hydrogen atom. An electron confined to such a small region *must* move rapidly with at least this speed and this energy. In Chapter 16, we will see that this subtle quantum effect is responsible for supporting white dwarf and neutron stars against the tremendous inward pull of gravity.

Quantum Mechanical Tunneling

When a ray of light attempts to travel from a glass prism into air, it may undergo *total internal reflection* if it strikes the surface at an angle greater than the critical angle θ_c , where the critical angle is related to the indices of refraction of the glass and air by

$$\sin \theta_c = \frac{n_{\text{air}}}{n_{\text{glass}}}.$$

This familiar result is nonetheless surprising because, even though the ray of light is totally reflected, the index of refraction of the outside air appears in this formula. In fact, the electromagnetic wave does enter the air, but it ceases to be oscillatory and instead dies away exponentially. In general, when a classical wave such as a water or light wave enters a

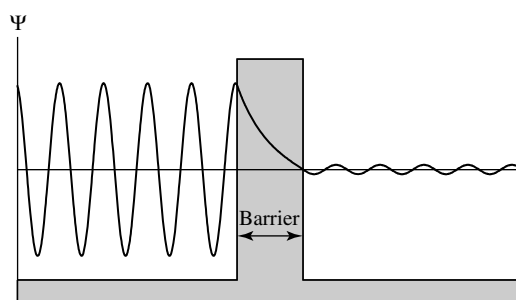


FIGURE 5.11 Quantum mechanical tunneling (barrier penetration) of a particle traveling to the right.

medium through which it cannot propagate, it becomes *evanescent* and its amplitude decays exponentially with distance.

This total internal reflection can in fact be frustrated by placing another prism next to the first prism so that their surfaces nearly (but not quite) touch. Then the evanescent wave in the air may enter the second prism before its amplitude has completely died away. The electromagnetic wave once again becomes oscillatory upon entering the glass, and so the ray of light has traveled from one prism to another without passing through the air gap between the prisms. In the language of particles, photons have **tunneled** from one prism to another without traveling in the space between them.

The wave–particle duality of nature implies that particles can also tunnel through a region of space (a barrier) in which they cannot exist classically, as illustrated in Fig. 5.11. The barrier must not be too wide (not more than a few particle wavelengths) if tunneling is to take place; otherwise, the amplitude of the evanescent wave will have declined to nearly zero. This is consistent with Heisenberg’s uncertainty principle, which implies that a particle’s location cannot be determined with an uncertainty that is less than its wavelength. Thus, if the barrier is only a few wavelengths wide, the particle may suddenly appear on the other side of the barrier. **Barrier penetration** is extremely important in radioactive decay, where alpha particles tunnel out of an atom’s nucleus; in modern electronics, where it is the basis for the “tunnel diode”; and inside stars, where the rates of nuclear fusion reactions depend upon tunneling.

Schrödinger’s Equation and the Quantum Mechanical Atom

What are the implications for Bohr’s model of the hydrogen atom? Heisenberg’s uncertainty principle does not allow classical orbits, with their simultaneously precise values of the electron’s position and momentum. Instead, the electron *orbitals* must be imagined as fuzzy clouds of probability, with the clouds being more “dense” in regions where the electron is more likely to be found (see Fig. 5.12). In 1925 a complete break from classical physics was imminent, one that would fully incorporate de Broglie’s matter waves.

As described in Section 3.3, Maxwell’s equations of electricity and magnetism can be manipulated to produce a wave equation for the electromagnetic waves that describe the propagation of photons. Similarly, a wave equation discovered in 1926 by Erwin Schrödinger (1877–1961), an Austrian physicist, led to a true **quantum mechanics**, the quantum analog of the classical mechanics that originated with Galileo and Newton. The Schrödinger

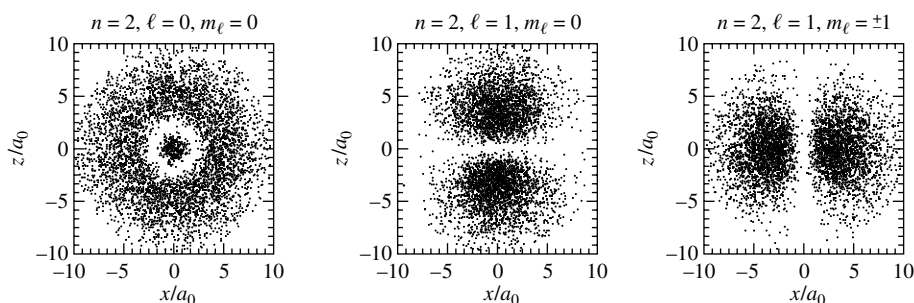


FIGURE 5.12 Electron orbitals of the hydrogen atom. Left: $2s$ orbital. Middle: $2p$ orbital with $m_\ell = 0$. Right: $2p$ orbital with $m_\ell = \pm 1$. The quantum numbers n , ℓ , and m_ℓ are described in the text.

equation can be solved for the probability waves that describe the allowed values of a particle's energy, momentum, and so on, as well as the particle's propagation through space. In particular, the Schrödinger equation can be solved analytically for the hydrogen atom, giving exactly the same set of allowed energies as those obtained by Bohr (cf. Eq. 5.11). However, in addition to the principal quantum number n , Schrödinger found that two additional quantum numbers, ℓ and m_ℓ , are required for a complete description of the electron orbitals. These additional numbers describe the angular momentum vector, $\mathbf{L}$, of the atom. Instead of the quantization used by Bohr, $L = n\hbar$, the solution to the Schrödinger equation shows that the permitted values of the magnitude of the angular momentum L are actually

$$L = \sqrt{\ell(\ell + 1)} \hbar, \quad (5.21)$$

where $\ell = 0, 1, 2, \dots, n - 1$, and n is the principal quantum number that determines the energy.

Note that it is common practice to refer to the angular momentum quantum numbers by their historical spectroscopic designations s, p, d, f, g, h , and so on, corresponding to $\ell = 0, 1, 2, 3, 4, 5$, etc. When the associated principle quantum number is used in combination with the angular momentum quantum number, the principle quantum number precedes the spectroscopic designation. For example, $(n = 2, \ell = 1)$ corresponds to $2p$, and $(n = 3, \ell = 2)$ is given as $3d$. This notation was used in the caption of Fig. 5.12 and is also used in Fig. 5.13.

The z -component of the angular momentum vector, L_z , can assume only the values $L_z = m_\ell \hbar$, with m_ℓ equal to any of the $2\ell + 1$ integers between $-\ell$ and $+\ell$ inclusive. Thus the angular momentum vector can point in $2\ell + 1$ different directions. For our purposes, the important point is that the values of the energy of an *isolated* hydrogen atom do not depend on ℓ and m_ℓ . In the absence of a preferred direction in space, the direction of the angular momentum has no effect on the atom's energy. Different orbitals, labeled by different values of ℓ and m_ℓ (see Fig. 5.12), are said to be **degenerate** if they have the same value of the principal quantum number n and so have the same energy. Electrons making a transition from a given orbital to one of several degenerate orbitals will produce the *same* spectral line, because they experience the same change in energy.

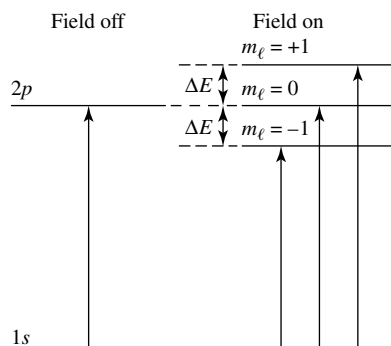


FIGURE 5.13 Splitting of absorption lines by the Zeeman effect.

However, the atom's surroundings may single out one spatial direction as being different from another. For example, an electron in an atom will feel the effect of an external magnetic field. The magnitude of this effect will depend on the $2\ell + 1$ possible orientations of the electron's motion, as given by m_ℓ , and the magnetic field strength, B , where the units of B are teslas (T).¹⁵ As the electron moves through the magnetic field, the normally degenerate orbitals acquire slightly different energies. Electrons making a transition between these formerly degenerate orbitals will thus produce spectral lines with slightly different frequencies. The splitting of spectral lines in a weak magnetic field is called the **Zeeman effect** and is shown in Fig. 5.13. The three frequencies of the split lines in the simplest case (called the *normal Zeeman effect*) are

$$\nu = \nu_0 \quad \text{and} \quad \nu_0 \pm \frac{eB}{4\pi\mu}, \quad (5.22)$$

where ν_0 is the frequency of the spectral line in the absence of a magnetic field and μ is the reduced mass. Although the energy levels are split into $2\ell + 1$ components, electron transitions involving these levels produce just three spectral lines with different polarizations.¹⁶ Viewed from different directions, it may happen that not all three lines will be visible. For example, when looking parallel to the magnetic field (as when looking down on a sunspot), the unshifted line of frequency ν_0 is absent.

Thus the Zeeman effect gives astronomers a probe of the magnetic fields observed around sunspots and on other stars. Even if the splitting of the spectral line is too small to be directly detected, the different polarizations across the closely spaced components can still be measured and the magnetic field strength deduced.

Example 5.4.3. Interstellar clouds may contain very weak magnetic fields, as small as $B \approx 2 \times 10^{-10}$ T. Nevertheless, astronomers have been able to measure this magnetic field. Using radio telescopes, they detect the variation in polarization that occurs across the

¹⁵Another commonly used unit of magnetic field strength is gauss, where $1 \text{ G} = 10^{-4} \text{ T}$. Earth's magnetic field is roughly 0.5 G , or $5 \times 10^{-5} \text{ T}$.

¹⁶See the discussion beginning on page 136 concerning selection rules.

blended Zeeman components of the absorption lines that are produced by these interstellar clouds of hydrogen gas. The change in frequency, $\Delta\nu$, produced by a magnetic field of this magnitude can be calculated from Eq. (5.22) by using the mass of the electron, m_e , for the reduced mass μ :

$$\Delta\nu = \frac{eB}{4\pi m_e} = 2.8 \text{ Hz},$$

a minute change. The total difference in frequency from one side of this blended line to the other is twice this amount, or 5.6 Hz. For comparison, the frequency of the radio wave emitted by hydrogen with $\lambda = 21 \text{ cm}$ is $\nu = c/\lambda = 1.4 \times 10^9 \text{ Hz}$, 250 million times larger!

Spin and the Pauli Exclusion Principle

Attempts to understand more complicated patterns of magnetic field splitting (the *anomalous Zeeman effect*), usually involving an even number of unequally spaced spectral lines, led physicists in 1925 to discover a *fourth* quantum number. In addition to its orbital motion, the electron possesses a **spin**. This is *not* a classical top-like rotation but purely a quantum effect that endows the electron with a *spin angular momentum* $\mathbf{S}$. $\mathbf{S}$ is a vector of constant magnitude

$$S = \sqrt{\frac{1}{2} \left(\frac{1}{2} + 1 \right)} \hbar = \frac{\sqrt{3}}{2} \hbar,$$

with a z -component $S_z = m_s \hbar$. The only values of the fourth quantum number, m_s , are $\pm \frac{1}{2}$.

With each orbital, or *quantum state*, labeled by four quantum numbers, physicists wondered how many electrons in a multielectron atom could occupy the same quantum state. The answer was supplied in 1925 by an Austrian theoretical physicist, Wolfgang Pauli (1900–1958): No two electrons can occupy the same quantum state. The **Pauli exclusion principle**, that *no two electrons can share the same set of four quantum numbers*, explained the electronic structure of atoms, thereby providing an explanation of the properties of the periodic table of the elements, the well-known chart from any introductory chemistry text. Despite this success, Pauli was unhappy about the lack of a firm theoretical understanding of electron spin. Spin was stitched onto quantum theory in an ad hoc manner, and the seams showed. Pauli lamented this patchwork theory and asked, “How can one avoid despondency if one thinks of the anomalous Zeeman effect?”

The final synthesis arrived in 1928 from an unexpected source. A brilliant English theoretical physicist, Paul Adrien Maurice Dirac (1902–1984), was working at Cambridge to combine Schrödinger’s wave equation with Einstein’s theory of special relativity. When he finally succeeded in writing a relativistic wave equation for the electron, he was delighted to see that the mathematical solution automatically included the spin of the electron. It also explained and extended the Pauli exclusion principle by dividing the world of particles into two fundamental groups: fermions and bosons. **Fermions**¹⁷ are particles such as electrons,

¹⁷The fermion is named after the Italian physicist Enrico Fermi (1901–1954).

protons, and neutrons¹⁸ that have a spin of $\frac{1}{2}\hbar$ (or an odd integer times $\frac{1}{2}\hbar$, such as $\frac{3}{2}\hbar$, $\frac{5}{2}\hbar$, . . .). Fermions obey the Pauli exclusion principle, so no two fermions of the same type can have the same set of quantum numbers. The exclusion principle for fermions, along with Heisenberg's uncertainty relation, explains the structure of white dwarfs and neutron stars, as will be discussed in Chapter 16. **Bosons**¹⁹ are particles such as photons that have an integral spin of $0, \hbar, 2\hbar, 3\hbar, \dots$. Bosons do not obey the Pauli exclusion principle, so any number of bosons can occupy the same quantum state.

As a final bonus, the Dirac equation predicted the existence of antiparticles. A particle and its antiparticle are identical except for their opposite electric charges and magnetic moments. Pairs of particles and antiparticles may be created from the energy of gamma-ray photons (according to $E = mc^2$). Conversely, particle-antiparticle pairs may annihilate each other, with their mass converted back into the energy of two gamma-ray photons. As we will see in Section 17.3, pair creation and annihilation play a major role in the evaporation of black holes.

The Complex Spectra of Atoms

With the full list of four quantum numbers (n , ℓ , m_ℓ , and m_s) that describe the detailed state of each electron in an atom, the number of possible energy levels increases rapidly with the number of electrons. When we take into account the additional complications of external magnetic fields, and the electromagnetic interactions between the electrons themselves and between the electrons and the nucleus, the spectra can become very complicated indeed. Figure 5.14 shows some of the available energy levels for the two electrons in the neutral helium atom.²⁰ Imagine the complexity of the relatively abundant iron atom with its 26 electrons!

Although energy levels exist for electrons with various combinations of quantum numbers, it is not always easy for an electron to make a transition from one quantum state with a specific set of quantum numbers to another quantum state. In particular, Nature imposes a set of **selection rules** that restrict certain transitions. For example, a careful investigation of Fig. 5.14 will show that only transitions involving $\Delta\ell = \pm 1$ are shown (from 1P to 1S , or from 1F to 1D , for instance). These transitions are referred to as **allowed** transitions and can happen spontaneously on timescales of 10^{-8} s. On the other hand, transitions that do not satisfy the requirement that $\Delta\ell = \pm 1$ are known as **forbidden** transitions.

In the case of the Zeeman effect first discussed on page 134, it was pointed out that only three transitions could occur between the $1s$ and $2p$ energy levels (recall Fig. 5.13). This is because of another set of selection rules requiring that $\Delta m_\ell = 0$ or ± 1 and forbidding transitions between orbitals if both orbitals have $m_\ell = 0$.

Although forbidden transitions may occur, they require much longer times if they are to occur with any significant probability. Since collisions between atoms trigger transitions and can compete with spontaneous transitions, very low gas densities are required for measurable intensities to be observed from forbidden transitions. Such environments do exist in astronomy, such as in the diffuse interstellar medium or in the outer atmospheres

¹⁸The neutron was not discovered until 1932 by James Chadwick (1891–1974), the same year that the positron (antimatter electron) was discovered by Carl Anderson (1905–1991).

¹⁹The boson is named in honor of the Indian physicist S. N. Bose (1894–1974).

²⁰Figure 5.14 is known as a Grotrian diagram.

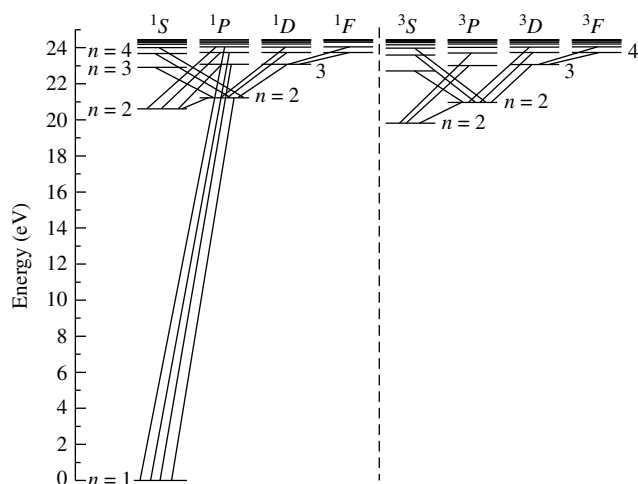


FIGURE 5.14 Some of the electronic energy levels of the helium atom. A small number of possible allowed transitions are also indicated. (Data courtesy of the National Institute of Standards and Technology.)

of stars. (It is beyond the scope of this text to discuss the detailed physics that underlies the existence of selection rules.)

The revolution in physics started by Max Planck culminated in the quantum atom and gave astronomers their most powerful tool: a theory that would enable them to analyze the spectral lines observed for stars, galaxies, and nebulae.²¹ Different atoms, and combinations of atoms in molecules, have orbitals of distinctly different energies; thus they can be identified by their spectral line “fingerprints.” The specific spectral lines produced by an atom or molecule depend on which orbitals are occupied by electrons. This, in turn, depends on its surroundings: the temperature, density, and pressure of its environment. These and other factors, such as the strength of a surrounding magnetic field, may be determined by a careful examination of spectral lines. Much of Chapters 8 and 9 will be devoted to the practical application of the quantum atom to stellar atmospheres.

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²¹Nearly all of the physicists mentioned in this chapter won the Nobel Prize for physics or chemistry in recognition of their work.

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PROBLEMS

- 5.1 Barnard's star, named after the American astronomer Edward E. Barnard (1857–1923), is an orange star in the constellation Ophiuchus. It has the largest known proper motion ($\mu = 10.3577'' \text{ yr}^{-1}$) and the fourth-largest parallax angle ($p = 0.54901''$). Only the stars in the triple system α Centauri have larger parallax angles. In the spectrum of Barnard's star, the $H\alpha$ absorption line is observed to have a wavelength of 656.034 nm when measured from the ground.
 - (a) Determine the radial velocity of Barnard's star.
 - (b) Determine the transverse velocity of Barnard's star.
 - (c) Calculate the speed of Barnard's star through space.
- 5.2 When salt is sprinkled on a flame, yellow light consisting of two closely spaced wavelengths, 588.997 nm and 589.594 nm, is produced. They are called the *sodium D lines* and were observed by Fraunhofer in the Sun's spectrum.
 - (a) If this light falls on a diffraction grating with 300 lines per millimeter, what is the angle between the second-order spectra of these two wavelengths?
 - (b) How many lines of this grating must be illuminated for the sodium D lines to just be resolved?
- 5.3 Show that $hc \simeq 1240 \text{ eV nm}$.
- 5.4 The photoelectric effect can be an important heating mechanism for the grains of dust found in interstellar clouds (see Section 12.1). The ejection of an electron leaves the grain with a positive charge, which affects the rates at which other electrons and ions collide with and stick to the grain to produce the heating. This process is particularly effective for ultraviolet photons ($\lambda \approx 100 \text{ nm}$) striking the smaller dust grains. If the average energy of the ejected electron is about 5 eV, estimate the work function of a typical dust grain.
- 5.5 Use Eq. (5.5) for the momentum of a photon, plus the conservation of relativistic momentum and energy [Eqs. (4.44) and (4.48), respectively], to derive Eq. (5.6) for the change in wavelength of the scattered photon in the Compton effect.

- 5.6 Consider the case of a “collision” between a photon and a free proton, initially at rest. What is the characteristic change in the wavelength of the scattered photon in units of nanometers? How does this compare with the Compton wavelength, λ_C ?
- 5.7 Verify that the units of Planck’s constant are the units of angular momentum.
- 5.8 A one-electron atom is an atom with Z protons in the nucleus and with all but one of its electrons lost to ionization.
- Starting with Coulomb’s law, determine expressions for the orbital radii and energies for a Bohr model of the one-electron atom with Z protons.
 - Find the radius of the ground-state orbit, the ground-state energy, and the ionization energy of singly ionized helium (He II).
 - Repeat part (b) for doubly ionized lithium (Li III).
- 5.9 To demonstrate the relative strengths of the electrical and gravitational forces of attraction between the electron and the proton in the Bohr atom, suppose the hydrogen atom were held together *solely* by the force of gravity. Determine the radius of the ground-state orbit (in units of nm and AU) and the energy of the ground state (in eV).
- 5.10 Calculate the energies and vacuum wavelengths of all possible photons that are emitted when the electron cascades from the $n = 3$ to the $n = 1$ orbit of the hydrogen atom.
- 5.11 Find the shortest vacuum-wavelength photon emitted by a downward electron transition in the Lyman, Balmer, and Paschen series. These wavelengths are known as the *series limits*. In which regions of the electromagnetic spectrum are these wavelengths found?
- 5.12 An electron in a television set reaches a speed of about $5 \times 10^7 \text{ m s}^{-1}$ before it hits the screen. What is the wavelength of this electron?
- 5.13 Consider the de Broglie wave of the electron in the Bohr atom. The circumference of the electron’s orbit must be an integral number of wavelengths, $n\lambda$; see Fig. 5.15. Otherwise, the electron wave will find itself *out of phase* and suffer destructive interference. Show that this requirement leads to Bohr’s condition for the quantization of angular momentum, Eq. (5.12).
- 5.14 A white dwarf is a very dense star, with its ions and electrons packed extremely close together. Each electron may be considered to be located within a region of size $\Delta x \approx 1.5 \times 10^{-12} \text{ m}$. Use Heisenberg’s uncertainty principle, Eq. (5.19), to estimate the minimum speed of the electron. Do you think that the effects of relativity will be important for these stars?
- 5.15 An electron spends roughly 10^{-8} s in the first excited state of the hydrogen atom before making a spontaneous downward transition to the ground state.

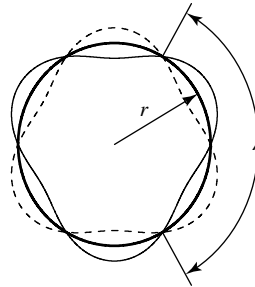


FIGURE 5.15 Three de Broglie wavelengths spanning an electron’s orbit in the Bohr atom.

- (a) Use Heisenberg's uncertainty principle (Eq. 5.20) to determine the uncertainty ΔE in the energy of the first excited state.
- (b) Calculate the uncertainty $\Delta\lambda$ in the wavelength of the photon involved in a transition (either upward or downward) between the ground and first excited states of the hydrogen atom. Why can you assume that $\Delta E = 0$ for the ground state?

This increase in the width of a spectral line is called *natural broadening*.

- 5.16** Each quantum state of the hydrogen atom is labeled by a set of four quantum numbers: $\{n, \ell, m_\ell, m_s\}$.
- (a) List the sets of quantum numbers for the hydrogen atom having $n = 1$, $n = 2$, and $n = 3$.
- (b) Show that the degeneracy of energy level n is $2n^2$.
- 5.17** The members of a class of stars known as Ap stars are distinguished by their strong global magnetic fields (usually a few tenths of one tesla).²² The star HD215441 has an unusually strong magnetic field of 3.4 T. Find the frequencies and wavelengths of the three components of the H α spectral line produced by the normal Zeeman effect for this magnetic field.

COMPUTER PROBLEM

- 5.18** One of the most important ideas of the physics of waves is that *any* complex waveform can be expressed as the sum of the harmonics of simple cosine and sine waves. That is, any wave function $f(x)$ can be written as

$$f(x) = a_0 + a_1 \cos x + a_2 \cos 2x + a_3 \cos 3x + a_4 \cos 4x + \cdots \\ + b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x + b_4 \sin 4x + \cdots$$

The coefficients a_n and b_n tell how much of each harmonic goes into the recipe for $f(x)$. This series of cosine and sine terms is called the *Fourier series* for $f(x)$. In general, both cosine and sine terms are needed, but in this problem you will use only the sine terms; all of the $a_n \equiv 0$.

On page 130, the process of constructing a wave pulse by adding a series of sine waves was described. The Fourier sine series that you will use to construct your wave employs only the *odd* harmonics and is given by

$$\Psi = \frac{2}{N+1} (\sin x - \sin 3x + \sin 5x - \sin 7x + \cdots \pm \sin Nx) = \frac{2}{N+1} \sum_{\substack{n=1 \\ n \text{ odd}}}^N (-1)^{(n-1)/2} \sin nx,$$

where N is an odd integer. The leading factor of $2/(N+1)$ does not change the shape of Ψ but scales the wave for convenience so that its maximum value is equal to 1 for any choice of N .

- (a) Graph Ψ for $N = 5$, using values of x (in radians) between 0 and π . What is the width, Δx , of the wave pulse?
- (b) Repeat part (a) for $N = 11$.
- (c) Repeat part (a) for $N = 21$.
- (d) Repeat part (a) for $N = 41$.
- (e) If Ψ represents the probability wave of a particle, for which value of N is the position of the particle known with the least uncertainty? For which value of N is the momentum of the particle known with the least uncertainty?

²²The letter A is the star's spectral type (to be discussed in Section 8.1), and the letter p stands for "peculiar."

CHAPTER

6

Telescopes

- 6.1 *Basic Optics*
- 6.2 *Optical Telescopes*
- 6.3 *Radio Telescopes*
- 6.4 *Infrared, Ultraviolet, X-ray, and Gamma-Ray Astronomy*
- 6.5 *All-Sky Surveys and Virtual Observatories*

6.1 ■ BASIC OPTICS

From the beginning, astronomy has been an observational science. In comparison with what was previously possible with the naked eye, Galileo's use of the new optical device known as the telescope greatly improved our ability to observe the universe (see Section 2.2). Today we continue to enhance our ability to "see" faint objects and to resolve them in greater detail. As a result, modern observational astronomy continues to supply scientists with more clues to the physical nature of our universe.

Although observational astronomy now covers the entire range of the electromagnetic spectrum, along with many areas of particle physics, the most familiar part of the field remains in the optical regime of the human eye (approximately 400 nm to 700 nm). Consequently, telescopes and detectors designed to investigate optical-wavelength radiation will be discussed in some detail. Furthermore, much of what we learn in studying telescopes and detectors in the optical regime will apply to other wavelength regions as well.

Refraction and Reflection

Galileo's telescope was a **refracting** telescope that made use of lenses through which light would pass, ultimately forming an image. Later, Newton designed and built a **reflecting** telescope that made use of mirrors as the principal optical component. Both refractors and reflectors remain in use today.

To understand the effects of an optical system on the light coming from an astronomical object, we will focus first on refracting telescopes. The path of a light ray through a lens can be understood using **Snell's law** of refraction. Recall that as a light ray travels from one transparent medium to another, its path is bent at the interface. The amount that the ray is bent depends on the ratio of the wavelength-dependent indices of refraction $n_\lambda \equiv c/v_\lambda$ of each material, where v_λ represents the speed of light within the specific medium.¹ If θ_1

¹It is only in a vacuum that $v_\lambda \equiv c$, independent of wavelength. The speed of light is wavelength-dependent in other environments.

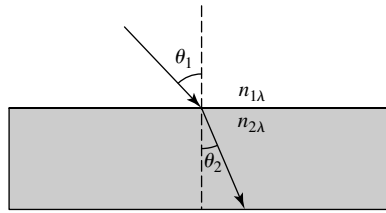
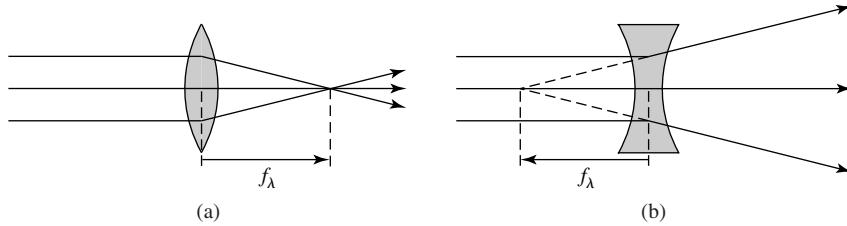


FIGURE 6.1 Snell's law of refraction.

FIGURE 6.2 (a) A converging lens, $f_\lambda > 0$. (b) A diverging lens, $f_\lambda < 0$.

is the angle of incidence, measured with respect to the normal to the interface between the two media, and θ_2 is the angle of refraction, also measured relative to the normal to the interface (see Fig. 6.1), then Snell's law is given by

$$n_{1\lambda} \sin \theta_1 = n_{2\lambda} \sin \theta_2. \quad (6.1)$$

If the surfaces of the lens are shaped properly, a beam of light rays of a given wavelength, originally traveling parallel to the axis of symmetry of the lens (called the **optical axis** of the system) can be brought to a focus at a point along that axis by a *converging* lens [Fig. 6.2(a)]. Alternatively, the light can be made to diverge by a *diverging* lens and the light rays will appear to originate from a single point along the axis [Fig. 6.2(b)]. The unique point in either case is referred to as the **focal point** of the lens, and the distance to that point from the center of the lens is known as the **focal length**, f . For a converging lens the focal length is taken to be positive, and for a diverging lens the focal length is negative.

The focal length of a given thin lens can be calculated directly from its index of refraction and geometry. If we assume that both surfaces of the lens are spheroidal, then it can be shown that the focal length f_λ is given by the **lensmaker's formula**,

$$\frac{1}{f_\lambda} = (n_\lambda - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right), \quad (6.2)$$

where n_λ is the index of refraction of the lens and R_1 and R_2 are the radii of curvature of each surface, taken to be positive if the specific surface is convex and negative if it is concave (see Fig. 6.3).²

²It is worth noting that many authors choose to define the sign convention for the radii of curvature in terms of the direction of the incident light. This choice means that Eq. (6.2) must be expressed in terms of the *difference* in the reciprocals of the radii of curvature.

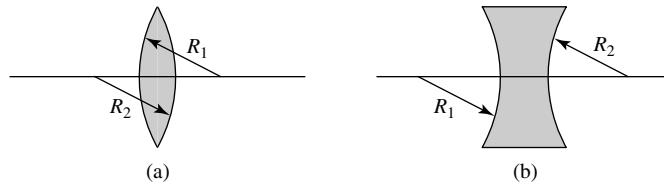


FIGURE 6.3 The sign convention for the radii of curvature of a lens in the lensmaker's formula. (a) $R_1 > 0$, $R_2 > 0$. (b) $R_1 < 0$, $R_2 < 0$.

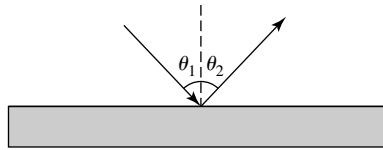


FIGURE 6.4 The law of reflection, $\theta_1 = \theta_2$.

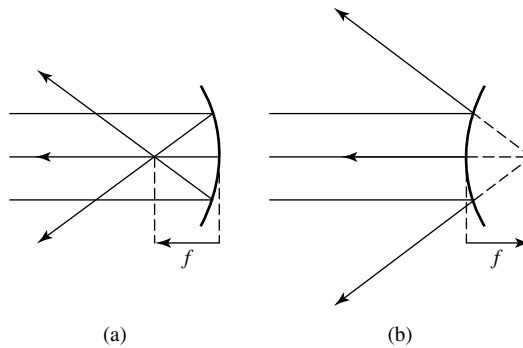


FIGURE 6.5 (a) A converging mirror, $f > 0$. (b) A diverging mirror, $f < 0$.

For mirrors f is wavelength-independent, since reflection depends only on the fact that the angle of incidence always equals the angle of reflection ($\theta_1 = \theta_2$; see Fig. 6.4). Furthermore, in the case of a spheroidal mirror (Fig. 6.5), the focal length becomes $f = R/2$, where R is the radius of curvature of the mirror, either positive (converging) or negative (diverging), a fact that can be demonstrated by simple geometry. Converging mirrors are generally used as the main mirrors in reflecting telescopes, although either diverging or flat mirrors may be used in other parts of the optical system.

The Focal Plane

For an extended object, the image will also necessarily be extended. If a photographic plate or some other detector is to record this image, the detector must be placed in the focal plane of the telescope. The **focal plane** is defined as the plane passing through the focal point and oriented perpendicular to the optical axis of the system. Since, for all practical purposes, any astronomical object can reasonably be assumed to be located infinitely far

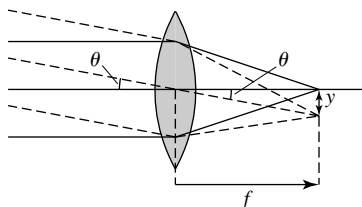


FIGURE 6.6 The plate scale, determined by the focal length of the optical system.

from the telescope,³ all of the rays coming from that object are essentially parallel to one another, although not necessarily parallel to the optical axis. If the rays are not parallel to the optical axis, distortion of the image can result; this is just one of many forms of *aberration* discussed later.

The image separation of two point sources on the focal plane is related to the focal length of the lens being used. Figure 6.6 shows the rays of two point sources, the direction of one source being along the optical axis of a converging lens and the other being at an angle θ with respect to the optical axis. At the position of the focal plane, the rays from the on-axis source will converge at the focal point while the rays from the other will *approximately* meet at a distance y from the focal point. Now, from simple geometry, y is given by

$$y = f \tan \theta$$

(the wavelength dependence of f is implicitly assumed). If it is assumed that the field of view of the telescope is small, then θ must also be small. Using the small-angle approximation, $\tan \theta \simeq \theta$, for θ expressed in radians, we find

$$y = f \theta. \quad (6.3)$$

This immediately leads to the differential relation known as the **plate scale**, $d\theta/dy$,

$$\boxed{\frac{d\theta}{dy} = \frac{1}{f}}, \quad (6.4)$$

which connects the angular separation of the objects with the linear separation of their images at the focal plane. As the focal length of the lens is increased, the linear separation of the images of two point sources separated by an angle θ also increases.

Resolution and the Rayleigh Criterion

Unfortunately, the ability to “see” two objects in space that have a small angular separation θ is not simply a matter of choosing a focal length sufficiently long to produce the necessary plate scale. A fundamental limit exists in our ability to *resolve* those objects. This limitation is due to diffraction produced by the advancing wavefronts of light coming from those objects.

³Technically, this implies that the distance to the astronomical object is much greater than the focal length of the telescope.

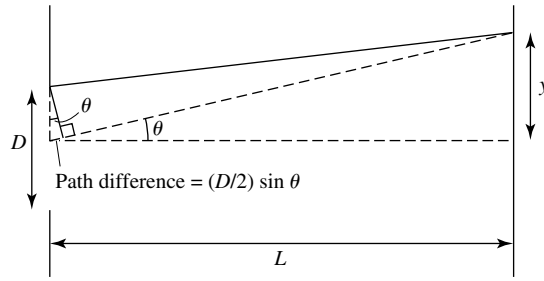


FIGURE 6.7 For a minimum to occur, the path difference between paired rays must be a half-wavelength.

This phenomenon is closely related to the well-known single-slit diffraction pattern, which is similar to the Young double-slit interference pattern discussed in Section 3.3.

To understand single-slit diffraction, consider a slit of width D (see Fig. 6.7). Assuming that the advancing wavefronts are coherent, any ray passing through the opening (or **aperture**) and arriving at a specific point in the focal plane can be thought of as being associated with another ray passing through the aperture exactly one-half of a slit width away and arriving at the same point. If the two rays are one-half wavelength ($\lambda/2$) out of phase, then destructive interference will occur. This leads to the relation

$$\frac{D}{2} \sin \theta = \frac{1}{2} \lambda,$$

or

$$\sin \theta = \frac{\lambda}{D}.$$

We can next consider dividing the aperture into four equal segments and pairing up a ray from the edge of the opening with one passing through a point one-quarter of a slit width away. For destructive interference to occur in this case, it is necessary that

$$\frac{D}{4} \sin \theta = \frac{1}{2} \lambda,$$

which gives

$$\sin \theta = 2 \frac{\lambda}{D}.$$

This analysis may be continued by considering the aperture as being divided into six segments, then eight segments, then ten segments, and so on. We see, therefore, that the condition for minima to occur as a result of destructive interference from light passing through a single slit is given in general by

$$\boxed{\sin \theta = m \frac{\lambda}{D}}, \quad (6.5)$$

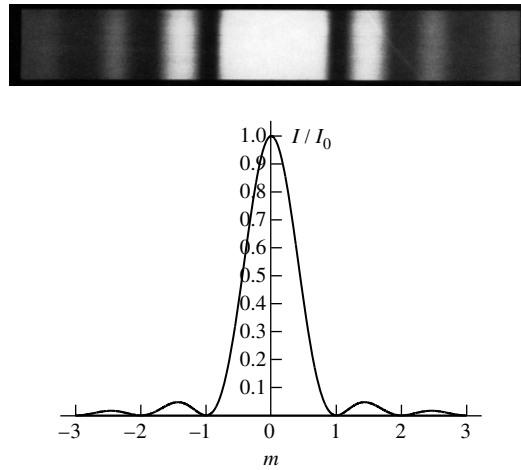


FIGURE 6.8 The diffraction pattern produced by a single slit. (Photograph from Cagnet, Francon, and Thierri, *Atlas of Optical Phenomena*, Springer-Verlag, Berlin, 1962.)

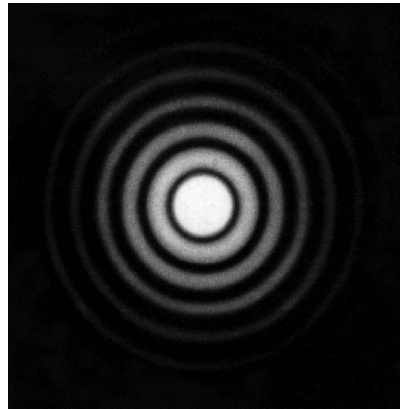


FIGURE 6.9 The circular aperture diffraction pattern of a point source. (Photograph from Cagnet, Francon, and Thierri, *Atlas of Optical Phenomena*, Springer-Verlag, Berlin, 1962.)

where $m = 1, 2, 3, \dots$ for dark fringes (as in Eq. 3.11). The intensity pattern produced by the light passing through a single slit is shown in Fig. 6.8.

The analysis for light passing through a circular aperture such as a telescope is similar, although somewhat more sophisticated. Due to the symmetry of the problem, the diffraction pattern appears as concentric rings (see Fig. 6.9). To evaluate this two-dimensional problem, it is necessary to perform a double integral over the aperture, considering the path differences of all possible pairs of rays passing through the aperture. The solution was first obtained in 1835 by Sir George Airy (1801–1892), Astronomer Royal of England; the central bright spot of the diffraction pattern is known as the **Airy disk**. Equation (6.5) remains appropriate for describing the locations of *both* the maxima and the minima, but m is no longer an integer.

TABLE 6.1 The locations and intensity maxima of the diffraction rings produced by a circular aperture.

Ring	m	I/I_0
Central maximum	0.000	1.00000
First minimum	1.220	
Second maximum	1.635	0.01750
Second minimum	2.233	
Third maximum	2.679	0.00416
Third minimum	3.238	

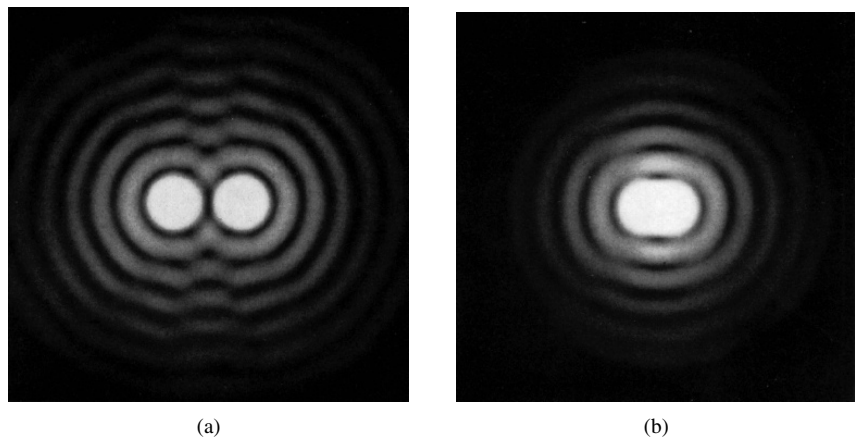
**FIGURE 6.10** The superimposed diffraction patterns from two point sources. (a) The sources are easily resolved. (b) The two sources are barely resolvable. (Photographs from Cagnet, Francon, and Thierri, *Atlas of Optical Phenomena*, Springer-Verlag, Berlin, 1962.)

Table 6.1 lists the values of m , along with the relative intensities of the maxima, for the first three orders.

As can be seen in Fig. 6.10, when the diffraction patterns of two sources are sufficiently close together (e.g., there is a very small angular separation, $\theta_{\min}$), the diffraction rings are no longer clearly distinguished and it becomes impossible to resolve the two sources. The two images are said to be unresolved when the central maximum of one pattern falls inside the location of the first minimum of the other. This *arbitrary* resolution condition is referred to as the **Rayleigh criterion**.⁴ Assuming that $\theta_{\min}$ is quite small, and invoking the small-angle approximation, $\sin \theta_{\min} \simeq \theta_{\min}$, where $\theta_{\min}$ is expressed in radians, the Rayleigh

⁴By undertaking a careful analysis of the diffraction patterns of the sources, it is possible to resolve objects that are somewhat more closely spaced than allowed by the Rayleigh criterion.

criterion is given by

$$\theta_{\min} = 1.22 \frac{\lambda}{D} \quad (6.6)$$

for a circular aperture. Therefore, the resolution of a telescope improves with increasing aperture size and when shorter wavelengths are observed, just as expected for diffraction phenomena.

Seeing

Unfortunately, despite the implications of Eq. (6.6), the resolution of ground-based optical telescopes does not improve without limit as the size of the primary lens or mirror is increased, unless certain complex, real-time adjustments are made to the optical system (see page 159). This consequence is due to the turbulent nature of Earth's atmosphere. Local changes in atmospheric temperature and density over distances ranging from centimeters to meters create regions where the light is refracted in nearly random directions, causing the image of a point source to become blurred. Since virtually all stars effectively appear as point sources, even when viewed through the largest telescopes, atmospheric turbulence produces the well-known “twinkling” of stellar images. The quality of the image of a stellar point source at a given observing location at a specific time is referred to as **seeing**. Some of the best seeing conditions found anywhere in the world are at **Mauna Kea Observatories** in Hawaii, located 4200 m (13,800 feet) above sea level, where the resolution is between 0.5'' and 0.6'' approximately 50% of the time, improving to 0.25'' on the best nights (Fig. 6.11). Other locations known for their excellent seeing are **Kitt Peak National Observatory** near Tucson, Arizona, **Tenerife** and **La Palma** of the Canary Islands, and several sites in the Chilean Andes Mountains [**Cerro Tololo Inter-American Observatory**, the **Cerro La Silla** and **Cerro Paranal** sites of the **European Southern Observatory**, and **Cerro Pachón**, location of Gemini South (Gemini North is on Mauna Kea)]. As a result, these sites have become locations where significant collections of optical telescopes and/or large-aperture optical telescopes have been built.

It is interesting to note that since the angular size of most planets is actually larger than the scale of atmospheric turbulence, distortions tend to be averaged out over the size of the image, and the “twinkling” effect is removed.

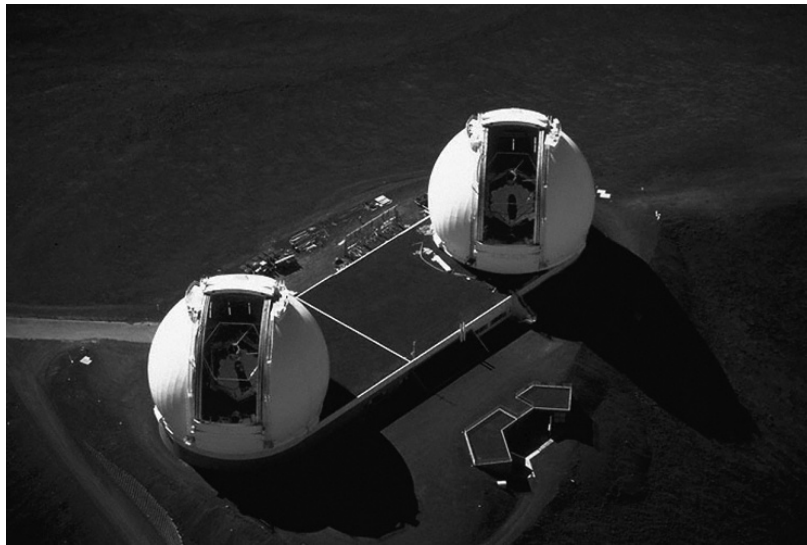
Example 6.1.1. After many years of delays, the **Hubble Space Telescope** (HST) was finally placed in an orbit 610 km (380 miles) high by the Space Shuttle *Discovery* in April 1990 [see Fig. 6.12(a)]. At this altitude, HST is above the obscuring atmosphere of Earth yet still accessible for needed repairs, instrument upgrades or replacement, or a boost in its constantly decaying orbit.⁵ HST is the most ambitious and, at a cost of approximately \$2 billion, the most expensive scientific project ever completed.

continued on page 150

⁵Decaying orbits are caused by the drag produced by Earth's extended, residual atmosphere. The extent of the atmosphere is determined in part by the heating associated with the solar cycle; see Section 11.3.



(a)



(b)

FIGURE 6.11 (a) The Mauna Kea Observatories in Hawaii. Among the telescopes visible in this view are Gemini North (open silver dome left of center, 8.1 m, optical/IR, operated by a seven-country consortium), the Canada-France-Hawaii Telescope (front center, 3.6 m optical), twin W. M. Keck Telescopes (back right, two 10 m, optical, Caltech and University of California, United States), and Japan's Subaru Telescope (left of Keck I and Keck II, 8.2 m, optical/IR). (Copyright 1998, Richard Wainscoat.) (b) Keck I and Keck II. These telescopes can be operated as an optical interferometer. (Copyright 1998, Richard Wainscoat.)

HST has a 2.4-m (94-inch) primary mirror. When we observe at the ultraviolet wavelength of the hydrogen Lyman alpha ($\text{Ly}\alpha$) line, 121.6 nm, the Rayleigh criterion implies a resolution limit of

$$\theta = 1.22 \left(\frac{121.6 \text{ nm}}{2.4 \text{ m}} \right) = 6.18 \times 10^{-8} \text{ rad} = 0.0127''.$$

This is roughly the equivalent of the angle subtended by a quarter from 400 km away! It was projected that HST would not quite be “diffraction-limited” in the ultraviolet region due to *extremely* small imperfections in the surfaces of the mirrors. Since resolution is proportional to wavelength and mirror defects become less significant as the wavelength increases, HST should have been nearly diffraction-limited at the red end of the visible spectrum. Unfortunately, because of an error in the grinding of the primary mirror, an optimal shape was not obtained. Consequently, those initial expectations were not realized until corrective optics packages were installed during a repair mission in December 1993 [Fig. 6.12(b)].

Aberrations

Both lens and mirror systems suffer from inherent image distortions known as **aberrations**. Often these aberrations are common to both types of systems, but **chromatic aberration** is unique to refracting telescopes. The problem stems from the fact that the focal length of a lens is wavelength-dependent. Equation (6.1) shows that since the index of refraction varies with wavelength, the angle of refraction at the interface between two different media must also depend on wavelength. This translates into a wavelength-dependent focal length (Eq. 6.2) and, as a result, a focal point for blue light that differs from that for red light. The problem of chromatic aberration can be diminished somewhat by the addition of correcting lenses. The demonstration of this procedure is left as an exercise.

Several aberrations result from the shape of the reflecting or refracting surface(s). Although it is easier, and therefore cheaper, to grind lenses and mirrors into spheroids, not all areas of these surfaces will focus a parallel set of light rays to a single point. This effect, known as **spherical aberration**, can be overcome by producing carefully designed optical surfaces (paraboloids).

The cause of HST’s initial imaging problems is a classic case of spherical aberration. A mistake that was made while grinding the primary left the center of the mirror too shallow by approximately two microns. The result of this minute error was that light reflected from near the edge of the mirror came to a focus almost 4 cm behind light reflected from the central portion. When the best possible compromise focal plane was used, the image of a point source (such as a distant star) had a definable central core and an extended, diffuse halo. Although the central core was quite small (0.1” radius), unfortunately it contained only 15% of the energy. The halo included more than half of the total energy and had a diameter of about 1.5” (typical of traditionally designed ground-based telescopes). The remainder of the energy (approximately 30%) was spread out over an even larger area. Some of HST’s original spherical aberration was compensated for by the use of computer programs designed to analyze the images produced by the flawed optical system and mathematically

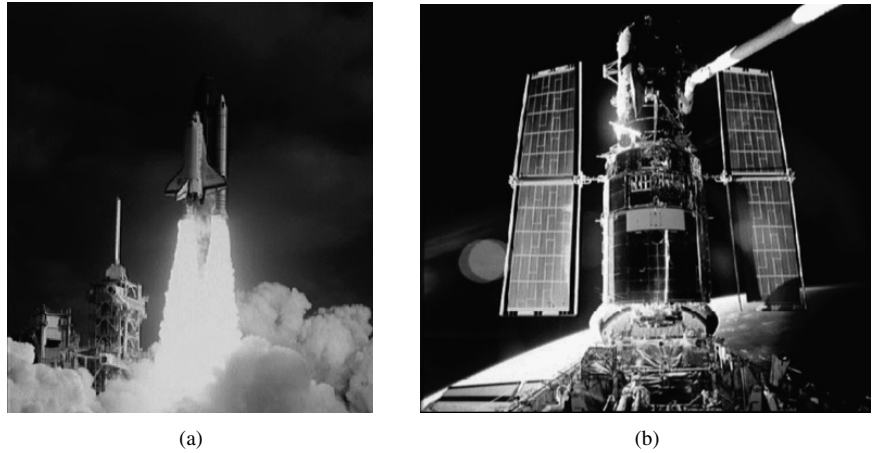


FIGURE 6.12 (a) The 1990 launch of the Hubble Space Telescope aboard the Space Shuttle *Discovery*. (b) HST and the Space Shuttle *Endeavour* during the December 1993 repair mission to install optical systems to compensate for a misshapen primary mirror. (Courtesy of NASA.)

create corrected versions. In addition, during the repair mission in 1993, special corrective optics packages were installed in the telescope. Today the spherical aberration problem of HST is only a bad memory of what can go wrong.

Even when paraboloids are used, mirrors are not necessarily free from aberrations. **Coma** produces elongated images of point sources that lie off the optical axis, because the focal lengths of paraboloids are a function of θ , the angle between the direction of an incoming light ray and the optical axis. **Astigmatism** is a defect that derives from having different parts of a lens or mirror converge an image at slightly different locations on the focal plane. When a lens or mirror is designed to correct for astigmatism, **curvature of field** can then be a problem. Curvature of field is due to the focusing of images on a curve rather than on a plane. Yet another potential difficulty occurs when the plate scale (Eq. 6.4) depends on the distance from the optical axis; this effect is referred to as **distortion of field**.

The Brightness of an Image

In addition to resolution and aberration issues, telescope design must also consider the desired **brightness** of an image. It might be assumed that the brightness of an extended (resolved) image would increase with the area of the telescope lens, since more photons are collected as the aperture size increases; however, this assumption is not necessarily correct. To understand the brightness of an image, we begin by considering the **intensity** of the radiation. Some of the energy radiated from an infinitesimal portion of the surface of the source of area $d\sigma$ [shown in Fig. 6.13(a)] will enter a cone of differential **solid angle** $d\Omega \equiv dA_{\perp}/r^2$, where $dA_{\perp}$ is an infinitesimal amount of surface area that is located a distance r from $d\sigma$ and oriented perpendicular to the position vector $\mathbf{r}$ [Fig. 6.13(b)].⁶ The

⁶The unit of solid angle is the steradian, sr. It is left as an exercise to show that $\Omega_{\text{tot}} = \oint d\Omega = 4\pi$ sr; the total solid angle about a point P , resulting from an integration over a closed surface containing that point, is 4π sr.

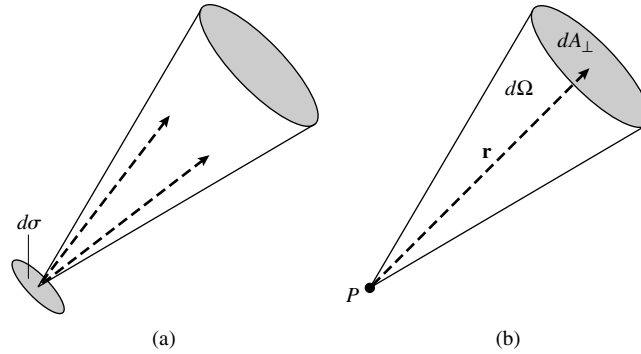


FIGURE 6.13 (a) The geometry of intensity. (b) The definition of solid angle.

intensity is given by the amount of energy per unit time interval dt , and per unit wavelength interval $d\lambda$, radiated from $d\sigma$ into a differential solid angle $d\Omega$; the units of intensity are $\text{W m}^{-2} \text{nm}^{-1} \text{sr}^{-1}$.

Consider an object located at a distance r far from a telescope of focal length f . Assuming that the object is effectively infinitely far away (i.e., $r \gg f$), the image intensity I_i may be determined from geometry. If an infinitesimal amount of surface area, dA_0 , of the object has a surface intensity given by I_0 , then the amount of energy per second per unit wavelength interval radiated into the solid angle defined by the telescope's aperture, $d\Omega_{T,0}$, is given by

$$I_0 d\Omega_{T,0} dA_0 = I_0 \left(\frac{A_T}{r^2} \right) dA_0,$$

where A_T is the area of the telescope's aperture [see Fig. 6.14(a)]. Since an image will form from the photons emitted by the object, all of the photons coming from dA_0 within the solid angle $d\Omega_{T,0}$ must strike an area dA_i on the focal plane.⁷ Therefore,

$$I_0 d\Omega_{T,0} dA_0 = I_i d\Omega_{T,i} dA_i,$$

where $d\Omega_{T,i}$ is the solid angle defined by the telescope's aperture as seen from the image, or

$$I_0 \left(\frac{A_T}{r^2} \right) dA_0 = I_i \left(\frac{A_T}{f^2} \right) dA_i.$$

Solving for the image intensity gives

$$I_i = I_0 \left(\frac{dA_0/r^2}{dA_i/f^2} \right).$$

However, as can be seen in Fig. 6.14(b), the solid angle $d\Omega_{0,T}$ containing the entire object as seen from the center of the telescope's aperture must equal the solid angle $d\Omega_{i,T}$ of the

⁷Assuming, of course, that no photons are absorbed or scattered out of the beam in transit.

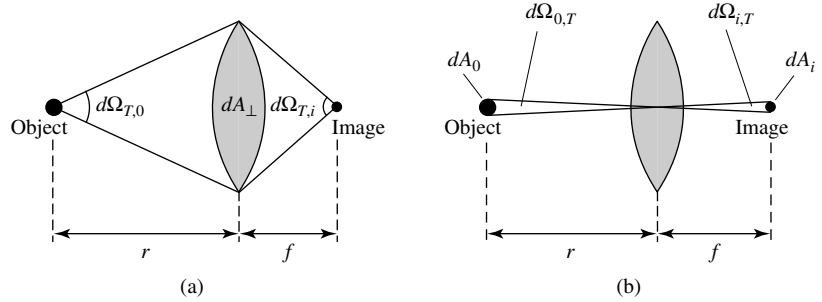


FIGURE 6.14 The effect of telescopes on image intensity ($r \gg f$). (a) The solid angles subtended by the telescope, as measured from the object and the image. (b) The solid angle subtended by the object and the image, as measured from the center of the telescope.

entire image, also seen from the telescope center, or $d\Omega_{0,T} = d\Omega_{i,T}$. This implies that

$$\frac{dA_0}{r^2} = \frac{dA_i}{f^2}.$$

Substituting into the expression for the image intensity gives the result that

$$I_i = I_0;$$

the image intensity is identical to the object intensity, independent of the area of the aperture. This result is completely analogous to the simple observation that a wall does not appear to get brighter when the observer walks toward it.

The concept that describes the effect of the light-gathering power of telescopes is the **illumination** J , the amount of light energy per second focused onto a unit area of the resolved image. Since the amount of light collected from the source is proportional to the area of the aperture, the illumination $J \propto \pi(D/2)^2 = \pi D^2/4$, where D is the diameter of the aperture. We have also shown that the linear size of the image is proportional to the focal length of the lens (Eq. 6.3); therefore, the image area must be proportional to f^2 , and correspondingly, the illumination must be inversely proportional to f^2 . Combining these results, the illumination must be proportional to the square of the ratio of the aperture diameter to the focal length. The inverse of this ratio is often referred to as the **focal ratio**,

$$F \equiv \frac{f}{D}. \quad (6.7)$$

Thus the illumination is related to the focal ratio by

$$J \propto \frac{1}{F^2}. \quad (6.8)$$

Since the number of photons per second striking a unit area of photographic plate or some other detector is described by the illumination, the illumination indicates the amount of time required to collect the photons needed to form a sufficiently bright image for analysis.

Example 6.1.2. The twin multimirror telescopes of the **Keck Observatory** at Mauna Kea have primary mirrors 10 m in diameter with focal lengths of 17.5 m. The focal ratios of these mirrors are

$$F = \frac{f}{D} = 1.75.$$

It is standard to express focal ratios in the form f/F , where $f/$ signifies that the focal ratio is being referenced. Using this notation, the Keck telescopes have 10-m, $f/1.75$ primary mirrors.

We now see that the size of the aperture of a telescope is critical for two reasons: A larger aperture both improves resolution and increases the illumination. On the other hand, a longer focal length increases the linear size of the image but decreases the illumination. For a *fixed* focal ratio, increasing the diameter of the telescope results in greater spatial resolution, but the illumination remains constant. The proper design of a telescope must take into account the principal applications that are intended for the instrument.

6.2 ■ OPTICAL TELESCOPES

In the last section we studied some of the fundamental aspects of optics in the context of astronomical observing. We now build on those concepts to consider design features of optical telescopes.

Refracting Telescopes

The major optical component of a refracting telescope is the primary or *objective* lens of focal length f_{obj} . The purpose of the objective lens is to collect as much light as possible and with the greatest possible resolution, bringing the light to a focus at the focal plane. A photographic plate or other detector may be placed at the focal plane to record the image, or the image may be viewed with an eyepiece, which serves as a magnifying glass. The eyepiece would be placed at a distance from the focal plane equal to its focal length, f_{eye} , causing the light rays to be refocused at infinity. Figure 6.15 shows the path of rays coming from a point source lying off the optical axis at an angle θ . The rays ultimately emerge from the eyepiece at an angle ϕ from the optical axis. The **angular magnification** produced by this arrangement of lenses can be shown to be (Problem 6.5)

$$m = \frac{f_{\text{obj}}}{f_{\text{eye}}}. \quad (6.9)$$

Clearly, eyepieces of different focal lengths can produce different angular magnifications. Viewing a large image requires a long objective focal length, in combination with a short focal length for the eyepiece.

Recall, however, that the illumination decreases with the square of the objective's focal length (Eq. 6.8). To compensate for the diminished illumination, a larger-diameter objective is needed. Unfortunately, significant practical limitations exist for the size of the objective

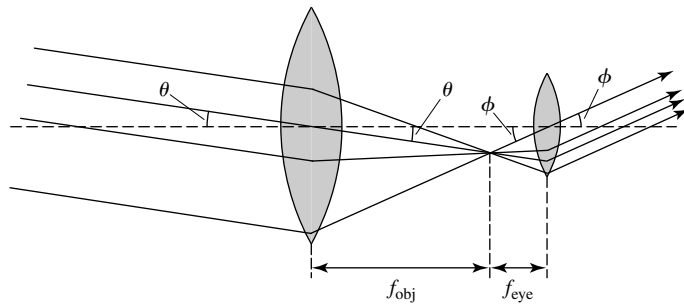


FIGURE 6.15 A refracting telescope is composed of an objective lens and an eyepiece.

lens of a refracting telescope. Because light must pass through the objective lens, it is possible to support the lens only from its edges. As a result, when the size and weight of the lens are increased, deformation in its shape occurs because of gravity. The specific form of the deformation depends on the position of the objective, which changes as the orientation of the telescope changes.

Another problem related to size is the difficulty in constructing a lens that is sufficiently free of defects. Since light must pass through the lens, its entire *volume* must be nearly optically perfect. Furthermore, *both* surfaces of the lens must be ground with great precision. Specifically, any defects in the material from which the lens is made and any deviations from the desired shape of the surface must be kept to less than some small fraction of the wavelength, typically $\lambda/20$. When observing at 500 nm, this implies that any defects must be smaller than approximately 25 nm. (Recall that the diameter of an atom is on the order of 0.1 nm.)

Yet another difficulty with a large objective lens occurs because of its slow thermal response. When the dome is opened, the temperature of the telescope must adjust to its new surroundings. This produces thermally driven air currents around the telescope, significantly affecting seeing. The shape of the telescope will also change as a consequence of thermal expansion, making it advantageous to minimize the “thermal mass” of the telescope as much as possible.

A mechanical problem also arises with long focal-length refractors. Due to the long lever arm involved, placing a massive detector on the end of the telescope will create a large amount of torque that requires compensation.

We have already discussed the unique problem of chromatic aberration in lenses, a complication not shared by mirrors. Considering all of the challenges inherent in the design and construction of refracting telescopes, the vast majority of all large modern telescopes are reflectors. The largest refracting telescope in use today is at the **Yerkes Observatory** in Williams Bay, Wisconsin (Fig. 6.16 on the following page). It was built in 1897 and has a 40-in (1.02-m) objective with a focal length of 19.36 m.

Reflecting Telescopes

With the exception of chromatic aberration, most of the basic optical principles already discussed apply equally well to reflectors and refractors. A reflecting telescope is designed by



FIGURE 6.16 The 40-in (1.02-m) telescope at Yerkes Observatory was built in 1897 and is the largest refractor in the world. (Courtesy of Yerkes Observatory.)

replacing the objective lens with a mirror, significantly reducing or completely eliminating many of the problems already discussed. Because the light does not pass through a mirror, only the *one* reflecting surface needs to be ground with precision. Also, the weight of the mirror can be minimized by creating a honeycomb structure behind the reflecting surface, removing a large amount of unnecessary mass. In fact, because the mirror is supported from behind rather than along its edges, it is possible to design an active system of pressure pads that can help to eliminate distortions in the mirror's shape produced by thermal effects and the changes in the gravitational force on the mirror as the telescope moves (a process known as **active optics**).

Reflecting telescopes are not completely free of drawbacks, however. Since the objective mirror reflects light back along the direction from which it came, the focal point of the mirror, known as the **prime focus**, is in the path of the incoming light [see Fig. 6.17(a)]. An observer or a detector can be placed at this position, but then some of the incident light is cut off (see Fig. 6.18). If the detector is too large, a substantial amount of light will be lost.

Isaac Newton first found a solution to the problem by placing a small, flat mirror in the reflected light's path, changing the location of the focal point; this arrangement is depicted in Fig. 6.17(b). Of course, the presence of this secondary mirror does block some of the incoming light from the primary, but if the ratio of the areas of the primary and secondary is sufficiently large, the effect of the lost light can be minimized. A **Newtonian** telescope design suffers from the drawback that the eyepiece (or detector) must be placed at a significant distance from the center of mass of the telescope. If a massive detector were used, it would exert a significant torque on the telescope.

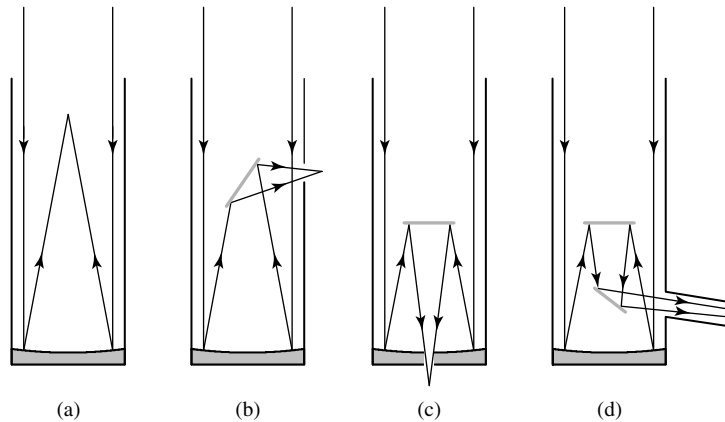


FIGURE 6.17 Schematic drawings of various telescope optical systems: (a) Prime focus, (b) Newtonian, (c) Cassegrain, (d) coudé.

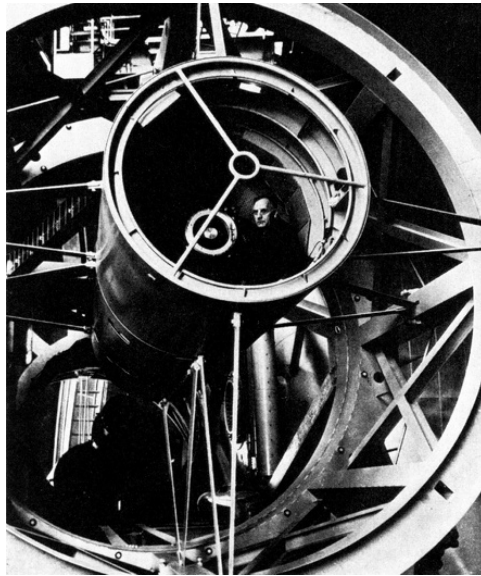


FIGURE 6.18 Edwin Hubble (1889–1953) working at the prime focus of the Hale reflecting telescope on Mount Palomar. (Courtesy of Palomar/Caltech.)

Since the region of the primary mirror located behind the secondary is effectively useless anyway, it is possible to bore a hole in the primary and use the secondary to reflect the light back through the hole. This **Cassegrain** design [Fig. 6.17(c)] makes it possible to place heavy instrument packages near the center of mass of the telescope and permits an observer to stay near the bottom of the telescope, rather than near the top, as is the case for Newtonians. In this type of design the secondary mirror is usually convex, effectively increasing the focal length of the system.

The classical Cassegrain design uses a parabolic primary mirror. However, an important modification to the Cassegrain design, known as a **Ritchey–Chrétien** design, uses a hyperbolic primary mirror rather than a parabolic one.

If the instrument package is too massive, it is often more effective to bring the light directly to a special laboratory in which the detector is located. A **coudé telescope** [Fig. 6.17(d)] uses a series of mirrors to reflect the light down the telescope's mount to a *coudé room* located below the telescope. Because of the extended optical path, it is possible to create a very long focal length with a coudé telescope. This can be particularly useful in high-resolution work or in high-dispersion spectral line studies (see Section 5.1).

A unique instrument is the **Schmidt** telescope, specifically designed to provide a wide-angle field of view with low distortion. Schmidt telescopes are generally used as cameras, with the photographic plate located at the prime focus. To minimize coma, a spheroidal primary mirror is used, combined with a “correcting” lens to help remove spherical aberration. Whereas a large Cassegrain telescope may have a field of view of a few arc minutes across, a Schmidt camera has a field of view of several degrees. These instruments provide important survey studies of large regions of the sky. For example, sky survey plates from the Palomar and UK Schmidt telescopes have been scanned to produce the Guide Star Catalogue II containing 998,402,801 objects as faint as 19.5 magnitudes.⁸ The stellar data in this catalog are being used to supply the reference (or *guide*) stars needed to orient the Hubble Space Telescope.

Telescope Mounts

Producing high-resolution, deep-sky images of faint objects requires that the telescope be pointed at a fixed region of the sky for an extended period of time. This is necessary so that enough photons will be collected to ensure that the desired object can be seen. Such *time integration* requires careful *guiding* (or positional control) of the telescope while compensating for the rotation of Earth.

In order to account for Earth's rotation, perhaps the most common type of telescope mount (especially for smaller telescopes) is the **equatorial mount**. It incorporates a polar axis that is aligned to the north celestial pole, and the telescope simply rotates about that axis to compensate for the changing altitude and azimuth of the object of interest. With an equatorial mount, it is a simple matter to adjust the position of the telescope in both right ascension and declination. Unfortunately, for a massive telescope an equatorial mount can be extremely expensive and difficult to build. An alternative, more easily constructed mount for large telescopes, the **altitude–azimuth mount**, permits motion both parallel and perpendicular to the horizon. In this case, however, the tracking of a celestial object requires the continuous calculation of its altitude and azimuth from knowledge of the object's right ascension and declination, combined with knowledge of the local sidereal time and latitude of the telescope. A second difficulty with altitude–azimuth mounts is the effect of the continuous rotation of image fields. Without proper adjustment, this can create complications when guiding the telescope during an extended exposure or when a spectrum is obtained by passing the light through a long slit. Fortunately, rapid computer calculations can compensate for all of these effects.

⁸The count of objects in GSC II quoted here was effective in May of 2006.

TABLE 6.2 Optical and/or near-infrared telescopes with apertures of 8 meters or more.

Name	Size (m)	Site	First Light
Gemini North	8.1	Mauna Kea, Hawaii	1999
Gemini South	8.1	Cerro Pachón, Chile	2002
Subaru	8.2	Mauna Kea, Hawaii	1999
Very Large Telescope (VLT)–Antu ^a	8.2	Cerro Paranal, Chile	1998
Very Large Telescope (VLT)–Kueyen ^a	8.2	Cerro Paranal, Chile	1999
Very Large Telescope (VLT)–Melipal ^a	8.2	Cerro Paranal, Chile	2000
Very Large Telescope (VLT)–Yepun ^a	8.2	Cerro Paranal, Chile	2000
Large Binocular Telescope (LBT) ^b	8.4 × 2	Mt. Graham, Arizona	2005
Hobby–Eberly Telescope (HET) ^c	9.2	McDonald Observatory, Texas	1999
Keck I ^d	10	Mauna Kea, Hawaii	1993
Keck II ^d	10	Mauna Kea, Hawaii	1996
Gran Telescopio Canarias (GTC)	10.4	La Palma, Canary Islands	2005
Southern African Large Telescope (SALT) ^e	11	Sutherland, South Africa	2005

^a The four 8.2-m VLT telescopes, together with three 1.8-m auxiliary telescopes, can serve as an optical/IR interferometer.

^b The two 8.4-m mirrors will sit on a single mount, with an effective collecting area of an 11.8-m aperture.

^c Mounted with a fixed altitude angle of 55°. The mirror measures 11.1 m by 9.8 m with an effective aperture of 9.2 m.

^d The two 10-m Keck telescopes, together with four 1.8-m outrigger telescopes, can serve as an optical/IR interferometer.

^e Mounted with a fixed altitude angle of 37°.

Large-Aperture Telescopes

In addition to long integration times, large aperture sizes play an important role in obtaining a sufficient number of photons to study a faint source (recall that the illumination is proportional to the diameter of the primary mirror of the telescope, Eq. 6.8). With tremendous improvements in telescope design, and aided by the development of high-speed computers, it has become possible to build very large-aperture telescopes. Table 6.2 contains a list of optical and/or near-infrared telescopes with apertures of greater than 8 m that are currently in operation. A number of much larger-aperture ground-based telescopes are also currently being considered, with effective mirror diameters ranging from 20 m to 100 m.

Adaptive Optics

While large-aperture ground-based telescopes are able to gather many more photons than smaller telescopes over the same time interval, they are generally unable to resolve the object any more effectively without significant effort. In fact, even a ground-based 10-m telescope located at a site with exceptional seeing (e.g. the Keck telescopes at Mauna Kea) cannot resolve a source any better than an amateur’s 20-cm backyard telescope can without the aid of *active optics* to correct distortions in the telescope’s mirrors (page 156) and **adaptive optics** to compensate for atmospheric turbulence. In the latter case, a small, deformable (“rubber”) mirror is employed that has tens or perhaps hundreds of piezoelectric crystals attached to the back that act like tiny actuators. In order to counteract changes in the shape of the wavefronts coming from the source due to

Earth's atmosphere, these crystals make micrometer-size adjustments to the shape of the mirror several hundred of times per second. In order to determine the changes that need to be applied, the telescope automatically monitors a guide star that is very near the target object.⁹ Fluctuations in the guide star determine the adjustments that must be made to the deformable mirror. This process is somewhat easier in the near-infrared simply because of the longer wave-lengths involved. As a result, adaptive optics systems have been successful in providing near-diffraction-limited images at near-infrared wavelengths.

Space-Based Observatories

In another effort to overcome the inherent imaging problems imposed by Earth's atmosphere, observational astronomy is also carried out in space. The **Hubble Space Telescope** (Fig. 6.12; named for Edwin Hubble) has a 2.4-m, $f/24$, primary that is the smoothest mirror ever constructed, with no surface imperfection larger than $1/50$ of the 632.8-nm test wavelength. Long exposures of 150 hours or more allow the telescope to "see" objects at least as faint as 30th magnitude. The optical system used by HST operates from 120 nm to 1 μm (ultraviolet to infrared, respectively) and is of the Ritchey–Chrétien type.

As HST approaches the end of its operational lifetime, plans are under way to replace it with the **James Webb Space Telescope** (JWST). The design specifications call for a telescope that will operate in the wavelength range between 600 nm and 28 μm , and it will have a primary mirror with a 6-m-diameter aperture. Unlike HST's low-Earth orbit, JWST will orbit about a gravitationally stable point that is located along the line connecting Earth and the Sun, but in the direction away from the Sun. This point, known as the second Lagrange point (L2), represents a balance between the gravitational forces of the Sun and Earth, and the centrifugal force due to its motion around the Sun as seen in a noninertial reference frame.¹⁰ This location was chosen for the spacecraft in order to minimize thermal emissions that could otherwise affect its infrared detectors.

Electronic Detectors

Although the human eye and photographic plates have traditionally been the tools of astronomers to record images and spectra, other, more efficient devices are typically used in modern astronomy today. In particular, the semiconductor detector known as the charge-coupled device (CCD) has revolutionized the way in which photons are counted. Whereas the human eye has a very low *quantum efficiency* of approximately 1% (one photon in one hundred is detected), and photographic plates do only slightly better, CCDs are able to detect nearly 100% of the incident photons. Moreover, CCDs are able to detect a very wide range of wavelengths. From soft (low-energy) X-rays to the infrared, they have a linear response: Ten times as many photons produce a signal ten times stronger. CCDs also have

⁹In most cases, a sufficiently bright guide star does not exist close enough to the target. At a small number of observatories, an artificial laser guide star may be used in these circumstances. This is accomplished by firing a very powerful and carefully tuned laser into the sky in order to excite sodium atoms at an altitude of approximately 90 km.

¹⁰For more information about the Lagrange points, see Figure 18.3 and the discussion in Section 18.1.

a wide dynamic range and so can differentiate between very bright and very dim objects that are viewed simultaneously.

A CCD works by collecting electrons that are excited into higher energy states (conduction bands) when the detector is struck by a photon (a process similar to the photoelectric effect). The number of electrons collected in each pixel is then proportional to the brightness of the image at that location. The $2\frac{1}{2}$ million pixels of HST's second-generation Wide Field and Planetary Camera (WF/PC 2) are the individual elements of four 800×800 pixel CCD cameras, with each pixel capable of holding up to 70,000 electrons. HST's Advanced Camera for Surveys (ACS) contains an array of 4144×4136 (or 17,139,584) pixels for high-resolution survey work.

Given the rapid improvement in both ground-based and orbital telescopes, along with the tremendous advancements in detector technologies, it is clear that the future of optical astronomy is indeed a bright one.

6.3 ■ RADIO TELESCOPES

In 1931 Karl Jansky (1905–1950) was conducting experiments for Bell Laboratories related to the production of radio-wavelength static from thunderstorms. During the course of his investigations Jansky discovered that some of the static in his receiver was of “extraterrestrial origin.” By 1935 he had correctly concluded that much of the signal he was measuring originated in the plane of the Milky Way, with the strongest emission coming from the constellation Sagittarius, which lies in the direction of the center of our Galaxy. Jansky's pioneering work represented the birth of *radio astronomy*, a whole new field of observational study.

Today radio astronomy plays an important role in our investigation of the electromagnetic spectrum. Radio waves are produced by a variety of mechanisms related to a range of physical processes, such as the interactions of charged particles with magnetic fields. This window on the universe provides astronomers and physicists with valuable clues to the inner workings of some of nature's most spectacular phenomena.

Spectral Flux Density

Since radio waves interact with matter differently than visible light does, the devices used to detect and measure it are necessarily very different from optical telescopes. The parabolic dish of a typical radio telescope reflects the radio energy of the source to an antenna. The signal is then amplified and processed to produce a *radio map* of the sky at a particular wavelength, like the one shown in Fig. 6.19.

The strength of a radio source is measured in terms of the **spectral flux density**, $S(\nu)$, the amount of energy per second, per unit frequency interval striking a unit area of the telescope. To determine the total amount of energy per second (the power) collected by the receiver, the spectral flux must be integrated over the telescope's collecting area and over the frequency interval for which the detector is sensitive, referred to as the *bandwidth*. If f_ν is a function describing the efficiency of the detector at the frequency ν , then the amount

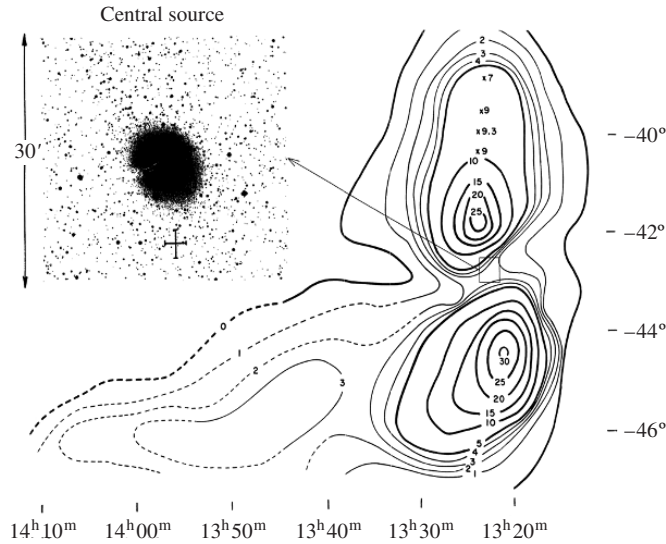


FIGURE 6.19 A radio map of Centaurus A, together with an optical image of the same region. The contours show lines of constant radio power. (Figure from Matthews, Morgan, and Schmidt, *Ap. J.*, 140, 35, 1964.)

of energy detected per second becomes¹¹

$$P = \int_A \int_{\nu} S(\nu) f_{\nu} d\nu dA. \quad (6.10)$$

If the detector is 100% efficient over a frequency interval $\Delta\nu$ (i.e., $f_{\nu} = 1$), and if $S(\nu)$ can be considered to be constant over that interval, then the integral simplifies to give

$$P = SA \Delta\nu,$$

where A is the effective area of the aperture.

A typical radio source has a spectral flux density $S(\nu)$ on the order of one jansky (Jy), where $1 \text{ Jy} = 10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}$. Spectral flux density measurements of several mJy are not uncommon. With such weak sources, a large aperture is needed to collect enough photons to be measurable.

Example 6.3.1. The third strongest radio source in the sky, after the Sun and Cassiopeia A (a nearby supernova remnant), is the galaxy Cygnus A (see Fig. 6.20). At 400 MHz (a wavelength of 75 cm), its spectral flux density is 4500 Jy. Assuming that a 25-m-diameter

¹¹A similar expression applies to optical telescopes since filters and detectors (including the human eye) are frequency dependent; see Section 3.6.

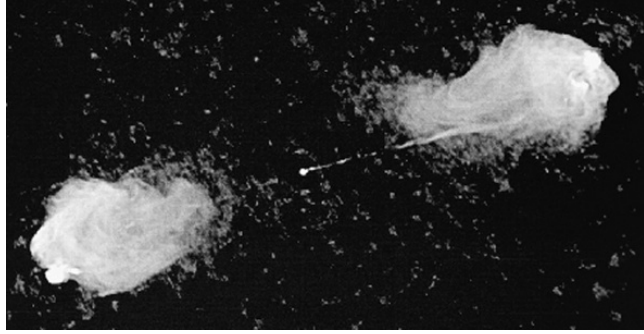


FIGURE 6.20 A Very Large Array (VLA; see page 166) radio-wavelength image of the relativistic jets coming from the core of the galaxy Cygnus A. (Courtesy of the National Radio Astronomy Observatory, ©NRAO/AUI.)

radio telescope is 100% efficient and is used to collect the radio energy of this source over a frequency bandwidth of 5 MHz, the total power detected by the receiver would be

$$P = S(\nu)\pi \left(\frac{D}{2}\right)^2 \Delta\nu = 1.1 \times 10^{-13} \text{ W.}$$

Improving Resolution: Large Apertures and Interferometry

One problem that radio telescopes share with optical telescopes is the need for greater resolution. Rayleigh's criterion (Eq. 6.6) applies to radio telescopes just as it does in the visible regime, except that radio wavelengths are much longer than those involved in optical work. Therefore, to obtain a level of resolution comparable to what is reached in the visible, much larger diameters are needed.

Example 6.3.2. To obtain a resolution of $1''$ at a wavelength of 21 cm using a single aperture, the dish diameter must be

$$D = 1.22 \frac{\lambda}{\theta} = 1.22 \left(\frac{21 \text{ cm}}{4.85 \times 10^{-6} \text{ rad}} \right) = 52.8 \text{ km.}$$

For comparison, the largest single-dish radio telescope in the world is the fixed dish 300 m (1000 ft) in diameter at **Arecibo Observatory**, Puerto Rico (see Fig. 6.21).

One advantage of working at such long wavelengths is that small deviations from an ideal parabolic shape are not nearly as crucial. Since the relevant criterion is to be within some small fraction of a wavelength (say $\lambda/20$) of what is considered a perfect shape, variations of 1 cm are tolerable when observing at 21 cm.

Although it is impractical to build individual dishes of sufficient size to produce the resolution at radio wavelengths that is anything like what is obtainable from the ground



FIGURE 6.21 The 300-m radio telescope at Arecibo Observatory, Puerto Rico. (Courtesy of the NAIC–Arecibo Observatory, which is operated by Cornell University for the National Science Foundation.)

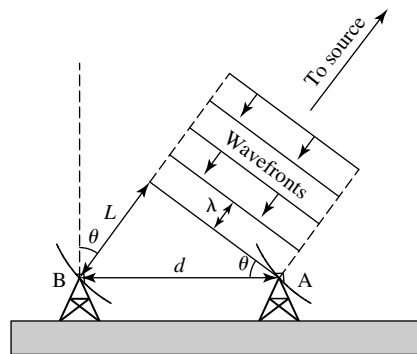


FIGURE 6.22 The technique of radio interferometry.

in the visible regime, astronomers have nevertheless been able to resolve radio images to better than $0.001''$. This remarkable resolution is accomplished through a process not unlike the interference technique used in the Young double-slit experiment.

Figure 6.22 shows two radio telescopes separated by a baseline of distance d . Since the distance from telescope B to the source is greater than the distance from telescope A to the source by an amount L , a specific wavefront will arrive at B after it has reached A. The two signals will be *in phase* and their superposition will result in a maximum if L is equal to an integral number of wavelengths ($L = n\lambda$, where $n = 0, 1, 2, \dots$ for constructive

interference). Similarly, if L is an odd integral number of half-wavelengths, then the signals will be exactly *out of phase* and a superposition of signals will result in a minimum in the signal strength [$L = (n - \frac{1}{2})\lambda$, where $n = 1, 2, \dots$ for destructive interference]. Since the pointing angle θ is related to d and L by

$$\sin \theta = \frac{L}{d}, \quad (6.11)$$

it is possible to accurately determine the position of the source by using the interference pattern produced by combining the signals of the two antennas. Equation (6.11) is completely analogous to Eq. (3.11) describing the Young double-slit experiment.

Clearly the ability to resolve an image improves with a longer baseline d . **Very long baseline interferometry** (VLBI) is possible over the size of a continent or even between continents. In such cases the data can be recorded on site and delivered to a central location for processing at a later time. It is only necessary that the observations be simultaneous and that the exact time of data acquisition be recorded.

Although a single antenna has its greatest level of sensitivity in the direction in which it is pointing, the antenna can also be sensitive to radio sources at angles far from the direction desired. Figure 6.23 shows a typical **antenna pattern** for a single radio telescope. It is a polar coordinate plot describing the direction of the antenna pattern along with the relative sensitivity in each direction; the longer the *lobe*, the more sensitive the telescope is in that direction. Two characteristics are immediately noticeable: First, the main lobe is not infinitesimally thin (the directionality of the beam is not perfect), and second, side lobes exist that can result in the accidental detection of unwanted sources that are indistinguishable from the desired source.

The narrowness of the main lobe is described by specifying its angular width at half its length, referred to as the **half-power beam width** (HPBW). This width can be decreased, and the effect of the side lobes can be significantly reduced, by the addition of other telescopes to produce the desired diffraction pattern. This property is analogous to the increase in sharpness of a grating diffraction pattern as the number of grating lines is increased; see Eq. (5.2).

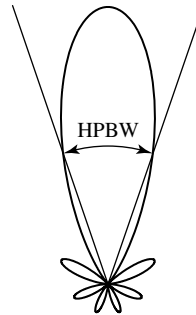


FIGURE 6.23 A typical antenna pattern for a single radio telescope. The width of the main lobe is describable by the half-power beam width (HPBW).



FIGURE 6.24 The Very Large Array (VLA) near Socorro, New Mexico. (Courtesy of the National Radio Astronomy Observatory, ©NRAO/AUI.)

The **Very Large Array** (VLA) located near Socorro, New Mexico, consists of 27 radio telescopes in a movable **Y** configuration with a maximum configuration diameter of 27 km. Each individual dish has a diameter of 25 m and uses receivers sensitive at a variety of frequencies (see Fig. 6.24). The signal from each of the separate telescopes is combined with all of the others and analyzed by computer to produce a high-resolution map of the sky; Fig. 6.20 is an example of an image produced by the VLA. Of course, along with the resolution gain, the 27 telescopes combine to produce an effective collecting area that is 27 times greater than that of an individual telescope.

The National Radio Astronomy Observatory (NRAO) plans to modernize and significantly expand the capabilities of the VLA. During Phase I, the **Expanded Very Large Array** (EVLA) will receive new, more sensitive receivers, extensive fiber-optic connections between the telescopes and the control facility, and greatly enhanced software and computational capabilities. Phase II of the expansion plan calls for the addition of approximately 8 new telescopes located throughout New Mexico that will augment the currently existing 27 telescopes. With baselines of up to 350 km, these new telescopes will greatly increase the resolution capability of the EVLA. The present VLA has a point source sensitivity of $10 \mu\text{Jy}$, a highest-frequency resolution of 381 Hz, and a spatial resolution (at 5 GHz) of $0.4''$. After completion of Phase II, the EVLA will have a point source sensitivity of $0.6 \mu\text{Jy}$, a highest-frequency resolution of 0.12 Hz, and a spatial resolution (at 5 GHz) of $0.04''$, which are improvements over the existing facility by one to two orders of magnitude.

NRAO also operates the **Very Long Baseline Array** (VLBA), composed of a series of 10 radio telescopes scattered throughout the continental United States, Hawaii, and St. Croix in the U.S. Virgin Islands. With a maximum baseline of 8600 km (5000 miles), the VLBA can achieve resolutions of better than $0.001''$.

In addition to these and other radio observatories around the world, a major international effort is under way to construct the **Atacama Large Millimeter Array (ALMA)**. ALMA will be composed of 50 12-m diameter antennas with baselines up to 12 km in length. Situated at an altitude of 5000 m (16,400 ft) in the Atacama desert region of Llano de Chajnantor in northern Chile, ALMA will be ideally located to work in the wavelength region from 10 mm to 350 μm (900 GHz to 70 GHz). At those wavelengths ALMA will be able to probe deeply into dusty regions of space where stars and planets are believed to be forming, as well as to study the earliest stages of galaxy formation—all critical problems in modern astrophysics. It is anticipated that ALMA will begin early scientific operations with a partial array in 2007. The full array is expected to be in operation by 2012.

6.4 ■ INFRARED, ULTRAVIOLET, X-RAY, AND GAMMA-RAY ASTRONOMY

Given the enormous amount of data supplied by optical and radio observations, it is natural to consider studies in other wavelength regions as well. Unfortunately, such observations are either difficult or impossible to perform from the ground because Earth's atmosphere is opaque to most wavelength regions outside of the visible and radio bands.

Atmospheric Windows in the Electromagnetic Spectrum

Figure 6.25 shows the transparency of the atmosphere as a function of wavelength. Long-wavelength ultraviolet radiation and some regions in the infrared are able to traverse the atmosphere with limited success, but other wavelength regions are completely blocked. For this reason, special measures must be taken to gather information at many photon energies.

The primary contributor to infrared absorption is water vapor. As a result, if an observatory can be placed above most of the atmospheric water vapor, some observations can be made from the ground. To this end, both NASA and the United Kingdom operate infrared telescopes (3 m and 3.8 m, respectively) on Mauna Kea where the humidity is quite low. However, even at an altitude of 4200 m, the problem is not completely solved. To get above more of the atmosphere, balloon and aircraft observations have also been used.

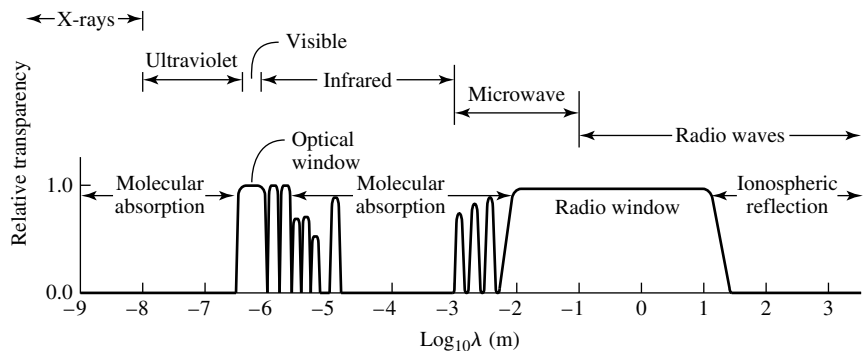


FIGURE 6.25 The transparency of Earth's atmosphere as a function of wavelength.

Besides atmospheric absorption, the situation in the infrared is complicated still further because steps must be taken to cool the detector, if not the entire telescope. Using Wien's displacement law (Eq. 3.15), the peak wavelength of a blackbody of temperature 300 K is found to be nearly $10\text{ }\mu\text{m}$. Thus the telescope and its detectors can produce radiation in just the wavelength region the observer might be interested in. Of course, the atmosphere itself can radiate in the infrared as well, including the production of molecular IR emission lines.

Observing Above the Atmosphere

In 1983 the **Infrared Astronomy Satellite** (IRAS) was placed in an orbit 900 km (560 miles) high, well above Earth's obscuring atmosphere. The 0.6-m imaging telescope was cooled to liquid helium temperatures, and its detectors were designed to observe at a variety of wavelengths from $12\text{ }\mu\text{m}$ to $100\text{ }\mu\text{m}$. Before its coolant was exhausted, IRAS proved to be very successful. Among its many accomplishments was the detection of dust in orbit around young stars, possibly indicating the formation of planetary systems. IRAS was also responsible for many important observations concerning the nature of galaxies.

Based upon the success of IRAS, the European Space Agency (in collaboration with Japan and the United States) launched the 0.6-m **Infrared Space Observatory** (ISO) in 1995. The observatory was cooled, just as IRAS was, but to obtain nearly 1000 times the resolution of IRAS, ISO was able to point toward a target for a much longer period of time, which enabled it to collect a greater number of photons. ISO ceased operations in 1998, after depletion of its liquid helium coolant.

The most recent and largest infrared observatory ever launched is the **Spitzer Space Telescope** [Fig. 6.26; named for Lyman Spitzer, Jr. (1914–1997)]. After many years of delay, this telescope was successfully placed into orbit on August 25, 2003. Trailing Earth in a heliocentric orbit, Spitzer is able to observe the heavens in the wavelength range from $3\text{ }\mu\text{m}$ to $180\text{ }\mu\text{m}$. With a 0.85-m, $f/12$ mirror constructed of light-weight beryllium and cooled to less than 5.5 K, the telescope is able to provide diffraction-limited performance at wavelengths of $6.5\text{ }\mu\text{m}$ and longer. It is anticipated that Spitzer will have an operational lifetime of at least 2.5 years.¹²

Designed to investigate the electromagnetic spectrum at the longer wavelengths of the microwave regime, the **COsmic Background Explorer** (COBE) was launched in 1989 and finally switched off in 1993. COBE made a number of important observations, including very precise measurements of the 2.7 K blackbody spectrum believed to be the remnant fireball of the Big Bang.

As in other wavelength regions, a number of challenges exist when observing in the ultraviolet portion of the electromagnetic spectrum. In this case, because of the short wavelengths involved (as compared to optical observations), great care must be taken to provide a very precise reflecting surface. As has already been mentioned, even the HST primary mirror has imperfections that prohibit shorter UV wavelengths from being observed at the theoretical resolution limit.

A second UV observing problem stems from the fact that glass is opaque to these short-wavelength photons (as it is for much of the infrared). Consequently, glass lenses cannot

¹²The Spitzer Space Telescope was the last of NASA's four great orbiting observatories to be placed in orbit. The others were the Hubble Space Telescope (Fig. 6.12), the Compton Gamma Ray Observatory [Fig. 6.27(a)], and the Chandra X-ray Observatory [Fig. 6.27(b)].

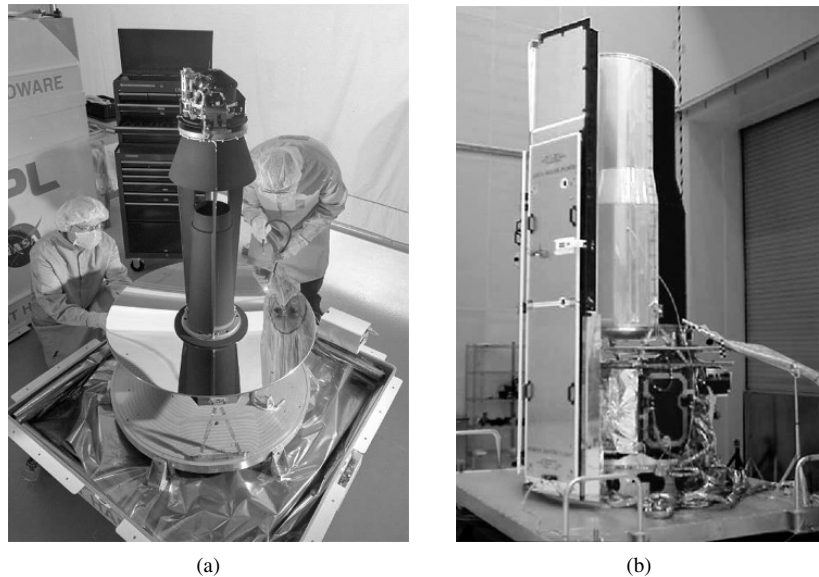


FIGURE 6.26 (a) The Spitzer Infrared Telescope during construction. The 0.85-m beryllium primary mirror is visible. (Courtesy of NASA/JPL.) (b) The almost fully assembled Spitzer Space Telescope in the laboratory. (Courtesy of NASA/JPL.)

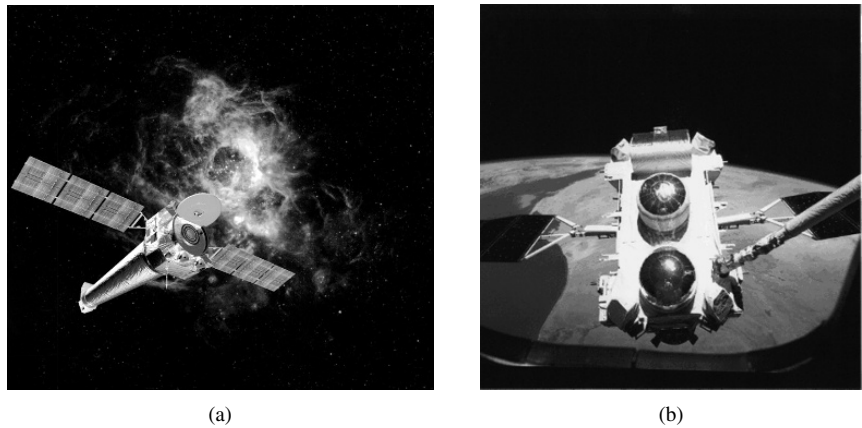


FIGURE 6.27 (a) An artist's conception of the Chandra X-ray Observatory. (Illustration: NASA/MSFC.) (b) The Compton Gamma Ray Observatory being deployed by the Space Shuttle *Atlantis* in 1991.

be used in the optical system of a telescope designed to observe in the ultraviolet. Lenses made of crystal provide an appropriate substitute, however.

A real workhorse of ultraviolet astronomy was the **International Ultraviolet Explorer**. Launched in 1978 and operational until 1996, the IUE proved to be a remarkably productive and durable instrument. Today, HST, with its sensitivity down to 120 nm, provides another important window on the ultraviolet universe. At even shorter wavelengths, the

Extreme Ultraviolet Explorer, launched in 1992 and switched off more than eight years later, made observations between 7 nm and 76 nm. The data from these telescopes have given astronomers important information concerning a vast array of astrophysical processes, including mass loss from hot stars, cataclysmic variable stars, and compact objects such as white dwarfs and pulsars.

At even shorter wavelengths, X-ray and gamma-ray astronomy yields information about very energetic phenomena, such as nuclear reaction processes and the environments around black holes. As a result of the very high photon energies involved, X-ray and gamma-ray observations require techniques that differ markedly from those at longer wavelengths. For instance, traditional glass mirrors are useless for forming images in this regime because of the great penetrating power of these photons. However, it is still possible to image sources by using grazing-incidence reflections (incident angle close to 90°). X-ray spectra can also be obtained using techniques such as Bragg scattering, an interference phenomenon produced by photon reflections from atoms in a regular crystal lattice. The distance between the atoms corresponds to the separation between slits in an optical diffraction grating.

In 1970 **UHURU** (also known as the Small Astronomy Satellite–1, SAS 1) made the first comprehensive survey of the X-ray sky. In the late 1970s the three **High Energy Astrophysical Observatories**, including the **Einstein Observatory**, discovered thousands of X-ray and gamma-ray sources. Between 1990 and 1999, the X-ray observatory ROSAT (the **Roentgen Satellite**), a German–American–British satellite consisting of two detectors and an imaging telescope operating in the range of 0.51 nm to 12.4 nm, investigated the hot coronas of stars, supernova remnants, and quasars. Japan’s **Advanced Satellite for Cosmology and Astrophysics**, which began its mission in 1993, also made valuable X-ray observations of the heavens before attitude control was lost as a result of a geomagnetic storm July 14, 2000.

Launched in 1999 and named for the Nobel Prize–winning astrophysicist Subrahmanyan Chandrasekhar (1910–1995), the **Chandra X-ray Observatory** [Fig. 6.27(a)] operates in the energy range from 0.2 keV to 10 keV (6.2 nm to 0.1 nm, respectively) with an angular resolution of approximately $0.5''$. Because X-rays cannot be focused in the same way that longer wavelengths can, grazing incidence mirrors are used to achieve the outstanding resolving power of Chandra.

The European Space Agency operates another X-ray telescope also launched in 1999, the **X-ray Multi-Mirror Newton Observatory** (XMM-Newton). Complementing Chandra’s sensitivity range, XMM-Newton operates between 0.01 nm and 1.2 nm.

The **Compton Gamma Ray Observatory** [CGRO; Fig. 6.27(b)] observed the heavens at wavelengths shorter than those measured by the X-ray telescopes. Placed into orbit by the Space Shuttle *Atlantis* in 1991, the observatory was deorbited into the Pacific Ocean in June 2000.

6.5 ■ ALL-SKY SURVEYS AND VIRTUAL OBSERVATORIES

Our ability to probe the heavens at wavelengths spanning the electromagnetic spectrum has provided an enormous amount of information not previously available from ground-based observations made exclusively in the visible wavelength regime. For example, Fig. 6.28 illustrates the change in the appearance of the sky when different wavelength regions are

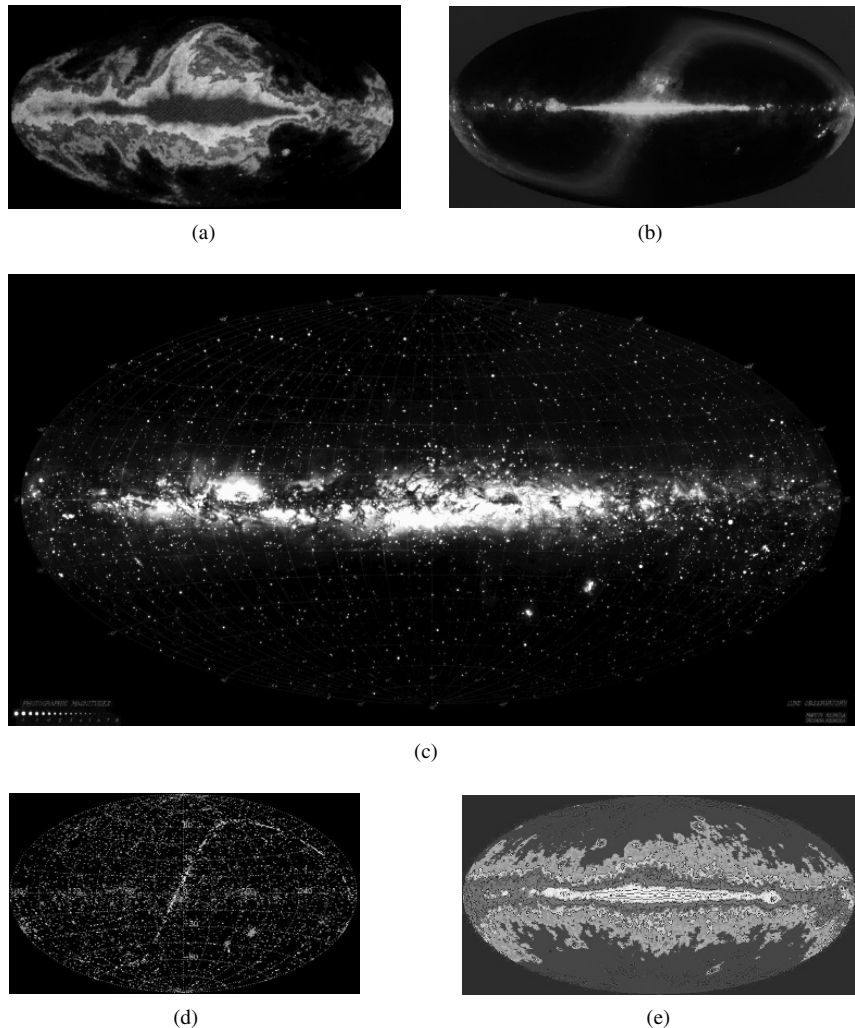


FIGURE 6.28 Observations of the entire sky as seen in various wavelengths. The plane of the Milky Way Galaxy is clearly evident running horizontally across each image. Also evident in several of the images is the plane of our Solar System, running diagonally from lower left to upper right. (a) Radio (courtesy of the Max-Planck Institut für Radioastronomie), (b) infrared (courtesy of the COBE Science Working Group and NASA's Goddard Space Flight Center), (c) visible (courtesy of Lund Observatory), (d) ultraviolet (courtesy of NASA's Goddard Space Flight Center), and (e) gamma-ray wavelengths (courtesy of NASA).

explored (radio, infrared, visible, ultraviolet, and gamma-rays). Notice that the plane of our Milky Way galaxy is clearly evident in each of the wavelength bands but that other features are not necessarily present in each image.

The ground-based and space-based observatories described in this chapter by no means constitute a complete list. For instance, along with the orbital telescopes discussed in Section 6.4 are many others designed for general observing, or to carry out specialized studies,

such as solar observations [e.g., the **Solar and Heliospheric Observatory** (SOHO) and the **Transition Region and Coronal Explorer** (TRACE)] or the determination of highly accurate positions of, and distances to, celestial objects¹³ [the **Hipparcos Space Astrometry Mission** (completed, ESA), the **SIM PlanetQuest Mission** (SIM, anticipated launch 2011, NASA), and **Gaia** (anticipated launch 2011, ESA)].

Moreover, from the ground, large-scale, automated surveys have been, or are being, conducted in various wavelength regimes. For instance, the visible **Sloan Digital Sky Survey** (SDSS) and the near infrared **Two-Micron All Sky Survey** (2MASS) will produce tremendous volumes of data that need to be analyzed. SDSS alone will result in 15 terabytes of data (comparable to all of the information contained in the Library of Congress). Petabyte-sized data sets are also being envisioned in the not-too-distant future.

Given the enormous volumes of data that have already been and will be produced by ground-based and space-based astronomical observatories, together with the tremendous amount of information that already exists in on-line journals and databases, great attention is being given to developing web-based virtual observatories. The goal of these projects is to create user interfaces that give astronomers access to already-existing observational data. For instance, an astronomer could query a virtual observatory database for all of the observations that have ever been made in a specified region of the sky over any wavelength band. That data would then be downloaded to the astronomer's desktop computer or mainframe for study. In order to accomplish this task, common data formats must be created, and data analysis and visualization tools must be developed to aide in this challenging project in information technology. At the time this text was written, several prototype virtual observatories had been developed, such as *Skyview*¹⁴ hosted by NASA's Goddard Space Flight Center, or the *Guide Star Catalogs* and the *Digitized Sky Survey*¹⁵ maintained by the Space Telescope Science Institute. On-line access to a large number of currently existing databases are also available at the National Space Science Data Center (NSSDC).¹⁶ In addition, several initiatives are under way to integrate and standardize the efforts. In the United States, the National Science Foundation (NSF) has funded the **National Virtual Observatory** project;¹⁷ in Europe, the **Astrophysical Virtual Observatory** project is under way; the United Kingdom is pursuing **Astrogrid**; and Australia is working on the **Australian Virtual Observatory**. It is hoped that all of these efforts will ultimately be combined to create an **International Virtual Observatory**.

With the past successes of ground-based and orbital observatories, astronomers have been able to make great strides in our understanding of the universe. Given the current advances in detectors, observational techniques, new observational facilities, and virtual observatories, the future holds tremendous promise for providing significantly improved studies of known objects in the heavens. However, perhaps the most exciting implications of these observational advances are to be found in as yet undiscovered and unanticipated phenomena.

¹³**Astrometry** is the subdiscipline in astronomy that determines positional information of celestial objects.

¹⁴Skyview is available at <http://skyview.gsfc.nasa.gov>.

¹⁵The Guide Star Catalogs and the Digitized Sky Survey can both be consulted at <http://www-gsss.stsci.edu>.

¹⁶The NSSDC website is located at <http://nssdc.gsfc.nasa.gov/>.

¹⁷The National Virtual Observatory website is located at <http://www.us-vo.org>.

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PROBLEMS

- 6.1** For some point P in space, show that for any arbitrary closed surface surrounding P , the integral over a solid angle about P gives

$$\Omega_{\text{tot}} = \oint d\Omega = 4\pi.$$

- 6.2** The light rays coming from an object do not, in general, travel parallel to the optical axis of a lens or mirror system. Consider an arrow to be the object, located a distance p from the center of a simple converging lens of focal length f , such that $p > f$. Assume that the arrow is perpendicular to the optical axis of the system with the tail of the arrow located on the axis. To locate the image, draw two light rays coming from the tip of the arrow:

- (i) One ray should follow a path *parallel* to the optical axis until it strikes the lens. It then bends toward the focal point of the side of the lens opposite the object.
- (ii) A second ray should pass directly through the center of the lens undeflected. (This assumes that the lens is sufficiently thin.)

The intersection of the two rays is the location of the tip of the image arrow. All other rays coming from the tip of the object that pass through the lens will also pass through the image tip. The tail of the image is located on the optical axis, a distance q from the center of the lens. The image should also be oriented perpendicular to the optical axis.

- (a) Using similar triangles, prove the relation

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}.$$

- (b) Show that if the distance of the object is much larger than the focal length of the lens ($p \gg f$), then the image is effectively located on the focal plane. This is essentially always the situation for astronomical observations.

The analysis of a diverging lens or a mirror (either converging or diverging) is similar and leads to the same relation between object distance, image distance, and focal length.

- 6.3** Show that if two lenses of focal lengths f_1 and f_2 can be considered to have zero physical separation, then the effective focal length of the combination of lenses is

$$\frac{1}{f_{\text{eff}}} = \frac{1}{f_1} + \frac{1}{f_2}.$$

Note: Assuming that the actual physical separation of the lenses is x , this approximation is strictly valid only when $f_1 \gg x$ and $f_2 \gg x$.

- 6.4** (a) Using the result of Problem 6.3, show that a compound lens system can be constructed from two lenses of different indices of refraction, $n_{1\lambda}$ and $n_{2\lambda}$, having the property that the resultant focal lengths of the compound lens at two specific wavelengths λ_1 and λ_2 , respectively, can be made equal, or

$$f_{\text{eff},\lambda_1} = f_{\text{eff},\lambda_2}.$$

- (b) Argue qualitatively that this condition does not guarantee that the focal length will be constant for all wavelengths.

- 6.5** Prove that the angular magnification of a telescope having an objective focal length of f_{obj} and an eyepiece focal length of f_{eye} is given by Eq. (6.9) when the objective and the eyepiece are separated by the sum of their focal lengths, $f_{\text{obj}} + f_{\text{eye}}$.

- 6.6** The diffraction pattern for a single slit (Figs. 6.7 and 6.8) is given by

$$I(\theta) = I_0 \left[\frac{\sin(\beta/2)}{\beta/2} \right]^2,$$

where $\beta \equiv 2\pi D \sin \theta / \lambda$.

- (a) Using l'Hôpital's rule, prove that the intensity at $\theta = 0$ is given by $I(0) = I_0$.
- (b) If the slit has an aperture of $1.0 \mu\text{m}$, what angle θ corresponds to the first minimum if the wavelength of the light is 500 nm ? Express your answer in degrees.
- 6.7** (a) Using the Rayleigh criterion, estimate the angular resolution limit of the human eye at 550 nm . Assume that the diameter of the pupil is 5 mm .
- (b) Compare your answer in part (a) to the angular diameters of the Moon and Jupiter. You may find the data in Appendix C helpful.
- (c) What can you conclude about the ability to resolve the Moon's disk and Jupiter's disk with the unaided eye?
- 6.8** (a) Using the Rayleigh criterion, estimate the theoretical diffraction limit for the angular resolution of a typical 20-cm (8-in) amateur telescope at 550 nm . Express your answer in arcseconds.
- (b) Using the information in Appendix C, estimate the minimum size of a crater on the Moon that can be resolved by a 20-cm (8-in) telescope.
- (c) Is this resolution limit likely to be achieved? Why or why not?
- 6.9** The New Technology Telescope (NTT) is operated by the European Southern Observatory at Cerro La Silla. This telescope was used as a testbed for evaluating the adaptive optics technology used in the VLT. The NTT has a 3.58-m primary mirror with a focal ratio of $f/2.2$.
- (a) Calculate the focal length of the primary mirror of the New Technology Telescope.
- (b) What is the value of the plate scale of the NTT?
- (c) ϵ Bootes is a double star system whose components are separated by $2.9''$. Calculate the linear separation of the images on the primary mirror focal plane of the NTT.
- 6.10** When operated in "planetary" mode, HST's WF/PC 2 has a focal ratio of $f/28.3$ with a plate scale of $0.0455'' \text{ pixel}^{-1}$. Estimate the angular size of the field of view of one CCD in the planetary mode.
- 6.11** Suppose that a radio telescope receiver has a bandwidth of 50 MHz centered at 1.430 GHz ($1 \text{ GHz} = 1000 \text{ MHz}$). Assume that, rather than being a perfect detector over the entire bandwidth, the receiver's frequency dependence is triangular, meaning that the sensitivity of the detector is 0% at the edges of the band and 100% at its center. This filter function can be expressed as

$$f_\nu = \begin{cases} \frac{\nu}{\nu_m - \nu_\ell} - \frac{\nu_\ell}{\nu_m - \nu_\ell} & \text{if } \nu_\ell \leq \nu \leq \nu_m \\ -\frac{\nu}{\nu_u - \nu_m} + \frac{\nu_u}{\nu_u - \nu_m} & \text{if } \nu_m \leq \nu \leq \nu_u \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Find the values of ν_ℓ , ν_m , and ν_u .
 - (b) Assume that the radio dish is a 100% efficient reflector over the receiver's bandwidth and has a diameter of 100 m. Assume also that the source NGC 2558 (a spiral galaxy with an apparent visual magnitude of 13.8) has a constant spectral flux density of $S = 2.5$ mJy over the detector bandwidth. Calculate the total power *measured* at the receiver.
 - (c) Estimate the power emitted at the source in this frequency range if $d = 100$ Mpc. Assume that the source emits the signal isotropically.
- 6.12** What would the diameter of a single radio dish need to be to have a collecting area equivalent to that of the 27 telescopes of the VLA?
- 6.13** How much must the pointing angle of a two-element radio interferometer be changed in order to move from one interference maximum to the next? Assume that the two telescopes are separated by the diameter of Earth and that the observation is being made at a wavelength of 21 cm. Express your answer in arcseconds.
- 6.14** Assuming that ALMA is completed with the currently envisioned 50 antennas, how many unique baselines will exist within the array?
- 6.15** The technical specifications for the planned SIM PlanetQuest mission call for the ability to resolve two point sources with an accuracy of better than $0.000004''$ for objects as faint as 20th magnitude in visible light. This will be accomplished through the use of optical interferometry.
- (a) Assuming that grass grows at the rate of 2 cm per week, and assuming that SIM could observe a blade of grass from a distance of 10 km, how long would it take for SIM to detect a measurable change in the length of the blade of grass?
 - (b) Using a baseline of the diameter of Earth's orbit, how far away will SIM be able to determine distances using trigonometric parallax, assuming the source is bright enough? (For reference, the distance from the Sun to the center of the Milky Way Galaxy is approximately 8 kpc.)
 - (c) From your answer to part (b), what would the apparent magnitude of the Sun be from that distance?
 - (d) The star Betelgeuse (in Orion) has an absolute magnitude of -5.14 . How far could Betelgeuse be from SIM and still be detected? (Neglect any effects of dust and gas between the star and the spacecraft.)
- 6.16** (a) Using data available in the text or on observatory websites, list the wavelength ranges (in m) and photon energy ranges (in eV) covered by the following telescopes: VLA, ALMA, Spitzer, JWST, VLT/VLTI, Keck/Keck Interferometer, HST, IUE, EUVE, Chandra, CGRO.
- (b) Graphically illustrate the wavelength coverage of each of the telescopes listed in part (a) by drawing a horizontal bar over a horizontal axis like the one shown in Fig. 6.25.
- (c) Using photon energies rather than wavelengths, create a graphic similar to the one in part (b).

COMPUTER PROBLEM

- 6.18** Suppose that two identical slits are situated next to each other in such a way that the axes of the slits are parallel and oriented vertically. Assume also that the two slits are the same distance from a flat screen. Different light sources of identical intensity are placed behind each slit so that the two sources are incoherent, which means that double-slit interference effects can be neglected.
- (a) If the two slits are separated by a distance such that the central maximum of the diffraction pattern corresponding to the first slit is located at the second minimum of the second slit's

diffraction pattern, plot the resulting superposition of intensities (i.e., the total intensity at each location). Include at least two minima to the left of the central maximum of the leftmost slit and at least two minima to the right of the central maximum of the rightmost slit. *Hint:* Refer to the equation given in Problem 6.6 and plot your results as a function of β .

- (b) Repeat your calculations for the case when the two slits are separated by a distance such that the central maximum of one slit falls at the location of the first minimum of the second (the Rayleigh criterion for single slits).
- (c) What can you conclude about the ability to resolve two individual sources (the slits) as the sources are brought progressively closer together?

PART II

The Nature of Stars

Binary Systems and Stellar Parameters

- 7.1 *The Classification of Binary Stars*
- 7.2 *Mass Determination Using Visual Binaries*
- 7.3 *Eclipsing, Spectroscopic Binaries*
- 7.4 *The Search for Extrasolar Planets*

7.1 ■ THE CLASSIFICATION OF BINARY STARS

A detailed understanding of the structure and evolution of stars (the goal of Part II) requires knowledge of their physical characteristics. We have seen that knowledge of blackbody radiation curves, spectra, and parallax enables us to determine a star's effective temperature, luminosity, radius, composition, and other parameters. However, the only direct way to determine the mass of a star is by studying its gravitational interaction with other objects.

In Chapter 2 Kepler's laws were used to calculate the masses of members of our Solar System. However, the universality of the gravitational force allows Kepler's laws to be generalized to include the orbits of stars about one another and even the orbital interactions of galaxies, as long as proper care is taken to refer all orbits to the center of mass of the system.

Fortunately, nature has provided ample opportunity for astronomers to observe binary star systems. At least half of all "stars" in the sky are actually multiple systems, two or more stars in orbit about a common center of mass. Analysis of the orbital parameters of these systems provides vital information about a variety of stellar characteristics, including mass.

The methods used to analyze the orbital data vary somewhat depending on the geometry of the system, its distance from the observer, and the relative masses and luminosities of the components. Consequently, binary star systems are classified according to their specific observational characteristics.

- **Optical double.** These systems are not actually binaries at all but simply two stars that lie along the same line of sight (i.e., they have similar right ascensions and declinations). As a consequence of their large physical separations, the stars are not gravitationally bound, and hence the system is not useful in determining stellar masses.
- **Visual binary.** Both stars in the binary can be resolved independently, and if the orbital period is not prohibitively long, it is possible to monitor the motion of each member of the system. These systems provide important information about the angular separation of the stars from their mutual center of mass. If the distance to the binary is also known, the linear separations of the stars can then be calculated.